WES 全流程分析路线图 · 矫正增强版

WES 数据分析实战路线图 · 矫正与增强版

Contents

About this edition · 关于本版 2
What each step contains · 每步包含什么 3
Issue registry at a glance · 问题清单速览 3
0. Environment, directories and data · 环境与数据 5
0.1 Required tools 5
0.2 Directory convention 5
0.3 Reference genome 6
0.4 Roadmap (and how we ran it on SLURM) 6
1. FASTQ files · 原始数据 7
1.1 Inspect FASTQ 7
2. Read quality control with fastp · 质控与预处理 8
2.1 Create QC output 8
2.2 Principle 原理 9
2.3 What happened 实跑结果 9
2.4 Pitfalls & fixes 坑与矫正 9
2.5 Interpretation 结果解读 9
2.6 Tuning & beyond 调优与扩展 10
3. Alignment with BWA-MEM · 比对 10
3.1 Principle 原理 10
3.2 SAM → BAM, sort, index (roadmap 3.2–3.3, verbatim) 10
3.3 Alignment metrics (roadmap 3.4, verbatim) 11
3.4 What happened 实跑结果 11
3.5 Tuning & beyond 调优与扩展 11
4. Mark PCR duplicates · 标记重复 12
4.1 Principle 原理 12
4.2 What happened 实跑结果 12
4.3 Pitfalls & fixes 坑与矫正 12
4.4 Tuning & beyond 调优与扩展 13
5. Base quality score recalibration (BQSR) · 碱基质量校准 13
5.1 Principle 原理 13
5.2 What happened 实跑结果 13
5.3 Pitfalls & fixes 坑与矫正 13
5.4 Tuning & beyond 调优与扩展 14
6. Germline variant calling · 胚系变异检测 14
6.1 Principle 原理 14
6.2 What happened 实跑结果 14
6.3 Pitfalls & fixes 坑与矫正 15
6.4 Tuning & beyond 调优与扩展 15
7. Variant quality score recalibration (VQSR) · 变异质量再校准 — the big correction 15
7.1 Principle 原理 15
7.2 What happened 实跑结果 — the two headline findings 15
7.3 The corrected workflow 矫正方案 17
7.4 Commands as we ran them 实跑命令说明 17

7.5 Interpretation 结果解读 18
8. Functional annotation with ANNOVAR · 功能注释 18
8.1 Principle 原理 18
8.2 Commands (roadmap 8.1–8.3, verbatim) 18
8.3 Pitfall (issue #2): the "richer protocol" block fails — and exactly why 19
8.4 Pitfall (issue #1 downstream): what the duplicated VCF does to ANNOVAR 19
8.5 What happened 实跑结果 19
8.6 Tuning & beyond 调优与扩展 20
9. Somatic TMB with matched normal (Mutect2) · 体细胞突变与 TMB 20
9.1 Principle 原理 20
9.2 Pitfalls & fixes 坑与矫正 20
9.3 What happened 实跑结果 21
9.4 Tuning & beyond 调优与扩展 22
10. CNV detection with CNVkit · 拷贝数变异 22
10.1 Principle 原理 22
10.2 Commands (roadmap 10.2, verbatim) and what it produced 22
10.3 Pitfalls & fixes 坑与矫正 (issue #4) 27
10.4 Interpretation 结果解读 28
10.5 Tuning & beyond 调优与扩展 28
11. Interpretation checklist · 结果解读检查清单 28
12. File summary · 产物清单 30
13. Summary · 总结与两轮对比 30
Reproduction · 复现说明 31
Appendix A. Issue registry · 问题清单（完整版） 32
A1. VQSR MergeVcfs duplication — **Major** 32
A2. ANNOVAR rich protocol names unavailable — **Major** 32
A3. VQSR on an undersized callset — **Major (statistical)** 33
A4. CNVkit `-m amplicon` on a hybrid-capture panel — **Major** 33
A5. Mutect2 without a Panel of Normals — Medium 33
A6. Dataset path inconsistency — Medium 33
A7. 33 GB intermediate SAM per sample — Medium (efficiency) 33
A8. TMB counts source inconsistency — Minor 33
A9. No on-target depth metric — Minor 33
A10. Naming inconsistencies — Minor 33
A11. Runtime environment quirks (not roadmap bugs, but you will hit them) — Info 33
A12. The teacher's reference result deviates from the printed roadmap — Major (comparability) 34
Appendix B. One-page corrected pipeline · 一页矫正版命令速查 34

About this edition · 关于本版

This document is a **corrected and heavily annotated edition** of the course roadmap "A Practical Roadmap for Whole-Exome Sequencing Data" (the original HTML/PDF remain in this directory for cross-reference). It was produced by **actually executing the original roadmap, command by command**, on a compute cluster — twice:

Round	Dataset	Depth over targets	Runtime	Purpose
Demo	`/lustre1/share/data_new` (18.2k / 25.8k	~0.5×	~25 min	Exercise the full 13-step pipeline mechanically

Round	Dataset	Depth over targets	Runtime	Purpose
	read pairs, OC / PBMC)			
Full	`/lustre1/share/data/OC_WES_all` (37.7M / 52.5M read pairs)	**measured 54.6× / 91.3×** usable on-target	~4 h (8 SLURM jobs, dependency-linked)	Produce real results, compare with the teacher's `analysis_full`

中文批注： 本版路线图的做法是——先把原版命令**原样**在集群上跑两轮（一轮 0.5× 降采样教学数据、一轮全量数据），把过程中遇到/发现的**每一个坑都记录在案**，再回头写这份矫正版。所以每个步骤里你都会看到"**原版怎么写 → 实际发生了什么 → 应该怎么改 → 为什么**"的完整链条。两轮实跑本身就是最好的对照组：0.5× 数据会暴露哪些步骤在"低深度下失效"，全量数据会暴露哪些步骤在"命令写法上就有问题"。

What each step contains · 每步包含什么

For every pipeline step, this edition adds six learning blocks around the original commands:

1. **Principle 原理** — what the algorithm actually does, in plain language.
2. **Commands** — the original roadmap commands (kept verbatim where they are correct; **corrections are shown explicitly and always justified**).
3. **What happened 实跑结果** — measured outputs from both runs (demo vs full).
4. **Pitfalls & fixes 坑与矫正** — every problem we hit, with evidence.
5. **Result interpretation 结果解读** — how to read the numbers; what "good" looks like.
6. **Tuning & beyond 调优与扩展** — parameter knobs, and what else the tool can do for you beyond this roadmap.

Issue registry at a glance · 问题清单速览

The full detailed registry is in the **Appendix A** section at the end of this document. Summary:

#	Severity	Where	Issue	Status in our runs
1	**Major**	§ 7.3	`MergeVcfs` of two full ApplyVQSR outputs **duplicates every variant** (each output contains all variants; merging them doubles the callset)	Confirmed in demo (14,782 → 29,564 records) **and in the teacher's own `analysis_full` output (204,178 → 408,356 records)**
2	**Major**	§ 8.3	Rich ANNOVAR protocol references databases that do not exist in the installed humandb (`dbnsfp30a`, `exac03`, `gnomad_genome`, `avsnp147` vs installed `dbnsfp54a`, `gnomad211_exome`, `gnomad41_exome`, `avsnp151`)	Confirmed failure in both runs
3	**Major**	§ 7	VQSR on a 2-sample, low-depth callset is statistically unreli-	Ran "successfully" on demo, but see the tranches analysis in § 7

#	Severity	Where	Issue	Status in our runs
			able even when it runs to completion	
4	**Major**	§ 10	`cnvkit.py batch -m amplicon` on a **hybrid-capture** panel (SureSelect V6) — wrong mode; also no gene names in BED, no access file	Ran, but produces genome-wide nonsense segments on demo (cn up to 6345)
5	Medium	§ 9	Mutect2 lacks a **Panel of Normals** (one exists on the cluster: `1000g_pon.hg38.vcf.gz`); germline resource uses 1000G high-conf instead of gnomAD AF-only	Runs, reduced filtering power
6	Medium	§ 0.2/ § 12	Data path confusion: README points to `OC_WES` (deprecated downsample), new roadmap points to `data_new`; old roadmap pointed to an **empty** directory	Three inconsistent references; this edition fixes them
7	Medium	§ 3	Literal roadmap writes a 33 GB SAM to disk before sorting — works, but wastes ~33 GB/sample and an extra hour	We piped `bwa | sort` in the full round (documented deviation)
8	Minor	§ 9	TMB counts PASS variants from `somatic.filtered.vcf.gz` while § 9.2 separately writes `somatic.pass.vcf.gz` (never used)	Cosmetic inconsistency
9	Minor	§ 4/ § 11	No on-target depth metric anywhere (flagstat is genome-wide) — the checklist asks "is coverage sufficient?" but no command measures it	This edition adds `CollectHsMetrics` (§ 11) — measured: OC 54.6×, PBMC 91.3×, demo 0.3×/0.5×
10	Minor	naming	`PBMC` vs `blood` naming mixed (RG `SM:blood`, dirs `PBMC`); teacher's own output dir is spelled `2_vairant_call`	Consistent-but-confusing; noted

#	Severity	Where	Issue	Status in our runs
11	Minor	§ 5	`BaseRecalibrator` runs whole-genome (no `-L`) — not wrong, ~2× slower for WES	Kept verbatim in both rounds
12	**Major***	reference	**The teacher's reference result itself deviates from the printed roadmap:** single-sample "joint" call (blood gVCF's `SM` tag = `OC`, so CombineGVCFs collapsed the pair) and **no `-L`** (72% of its 204,178 sites are off-target), plus issue #1's duplication (408,356 = 2×204,178). *Not a roadmap-text bug — an execution deviation in the reference artifacts	Read from the VCF headers; documented in Appendix A12 so callset comparisons are interpreted correctly

中文批注：最重要的是 #1（VQSR 合并导致每条变异重复两遍，连老师的参考结果都带着这个 bug）和 #4（CNVkit 模式选错）。这两个都不影响"管线能不能跑完"，但**直接影响结果文件能不能用**——这正是"跑通了"和"跑对了"的区别。

0. Environment, directories and data · 环境与数据

0.1 Required tools

Everything runs from the shared conda environment on the cluster:

```
source /lustre1/share/miniconda3/etc/profile.d/conda.sh
conda activate wes
```

Tool versions verified on a compute node (not the login node — see § 0.4): fastp 0.23.2, BWA 0.7.19-r1273, samtools, GATK 4.6.2.0 (Java), CNVkit 0.9.11, ANNOVAR (2020s build with hg38 humandb), bcftools.

中文批注：集群上所有重计算都必须 `sbatch` 到 `cn-long` 分区做，登录节点只用来编辑文件和提交作业。本文所有实跑结果都来自 SLURM 作业（见 § 13 复现说明）。

0.2 Directory convention

The roadmap's environment block, kept verbatim (this is the new version's only difference from the old one, which pointed `$DATA` at an **empty** directory — see issue #6):

```
export SHARE=/lustre1/share
export REF=$SHARE/references
export TARGETS=$REF/S07604514_AllTracks_V6_60_hg38.bed
export REFERENCE=$REF/hg38.fa
export DBSNP=$REF/dbsnp_138.hg38.vcf
export INDELS=$REF/Mills_and_1000G_gold_standard.indels.hg38.vcf
export HIGHCONF=$REF/1000G_phase1.snps.high_confidence.hg38.vcf
```

```
export HAPMAP=$REF/hapmap_3.3.hg38.vcf
export OMNI=$REF/1000G_omni2.5.hg38.vcf
export ANNOVAR=$SHARE/annovar_new
export HUMANDB=$ANNOVAR/humandb
export PATH=$ANNOVAR:$PATH
export DATA=$SHARE/data_new        # demo round; full round uses $SHARE/data/OC_WES_all
export BASE=$HOME/wes_run_cluster  # we used explicit run dirs, see §13
export OUT=$BASE
```

What lives where (verified by `ls`, not assumed):

Path	Content	Notes
`/lustre1/share/references/`	`hg38.fa` + BWA index + `.dict`/`.fai`; GATK bundle (`dbsnp_138`, Mills, 1000G high-conf, hapmap, omni) **with `.idx` present**; target BED `S07604514_AllTracks_V6_60_hg38.bed`; **`1000g_pon.hg38.vcf.gz` + `.tbi`** (a ready-made Panel of Normals!); `access-5kb.hg38.bed`	read-only share
`/lustre1/share/data_new/`	demo FASTQs: `OC/OC_R1.fq.gz` etc. (182k pairs), `PBMC/blood_R1.fq.gz` (258k pairs)	~0.48% subsample of the full data ≈ 0.5× over targets
`/lustre1/share/data/OC_WES_all/`	full FASTQs: OC 37.7M pairs (10.4 Gbp), PBMC 52.5M pairs (15.0 Gbp)	ideal ≈95×/≈135×; **measured usable 54.6× / 91.3×** (§11)
`/lustre1/share/data/OC_WES/`	older downsample (same size, **different content** — different md5)	deprecated; README still points here (issue #6)
`/lustre1/share/data/analysis_full/`	teacher's reference results (full data): alignment, joint VCF, VQSR, ANNOVAR	**only up to §8**; no somatic/TMB or CNVkit reference; `4_cnv/` is empty
`/lustre1/share/annovar_new/humandb/`	hg38 databases: `refGeneWithVer`, `cytoBand`, **`dbnsfp54a`, `gnomad211_exome`, `gnomad41_exome`, `avsnp151`**	the rich-protocol names in §8.3 of the roadmap are **not** here (issue #2)

Target region: the BED contains 243,359 intervals totaling **60.51 Mb** (this number matters twice: as `-L` intervals and as the TMB denominator in §9.3).

0.3 Reference genome

hg38 (`hg38.fa`), pre-indexed for BWA and GATK. **Check before you compute** (we verified all of these exist — a missing `.idx` is the single most common "first job fails" cause):

```
hg38.fa  hg38.fa.fai  hg38.dict
hg38.fa.{amb,ann,bwt,pac,sa}          # BWA index
dbsnp_138.hg38.vcf{,.idx}             # also needed by HaplotypeCaller -D
Mills_and_1000G_gold_standard.indels.hg38.vcf{,.idx}
1000G_phase1.snps.high_confidence.hg38.vcf{,.idx}
hapmap_3.3.hg38.vcf{,.idx}
1000G_omni2.5.hg38.vcf{,.idx}
```

0.4 Roadmap (and how we ran it on SLURM)

The roadmap's steps were executed as sbatch jobs on the `cn-long` partition (20-core nodes, 25 h wall limit). Three cluster-specific facts that cost us an hour to discover — record them once, reuse forever:

1. **Account/QoS:** jobs must specify `--account=bjx131_g1 --qos=bjx131cnl` (plain `--qos=bjx131cnl` without the account is rejected: "Invalid qos specification").

2. **Memory:** the `cn-long` nodes report `RealMemory=1` to SLURM, so any explicit `--mem=NN` is rejected ("Memory specification can not be satisfied"). Use **`--mem=0`** ("take the whole node's memory").
3. **conda + `set -u`:** if your script uses `set -u`, activate conda **before** turning it on — conda's `activate.d` scripts reference unbound variables and will kill the job in one second.

All reproduction scripts are in `scripts/` of each run directory (see § 13).

The two rounds at a glance (all jobs on `cn-long`, account `bjx131_g1`, QoS `bjx131cnl`, `--mem=0`):

Round	Job(s)	Steps	Wall time	Outcome
demo	3048	3→10 (verbatim, incl. expected-failure guards)	~25 min	all steps executed; deviations log: 1 (rich-protocol fail)
full	3049/3050	prep per sample: fastp → `bwa|sort` → index/flagstat → MarkDup → BQSR → HC-GVCF	2 h 20 m / 3 h 42 m (parallel)	both clean
full	3051	6.3→8 verbatim (joint → VQSR → MergeVcfs → ANNOVAR basic)	11 min	duplication bug reproduced (§ 7.2)
full	3052	9 verbatim (Mutect2 → orientation model → FilterMutectCalls → TMB)	1 h 13 m	TMB = 1.78 mut/Mb
full	3053	10 verbatim (`-m amplicon`)	5.6 min	runs; empty antitarget baseline (§ 10.2)
full	3054/3056→3061	**corrected extras**: CollectHsMetrics ×4 samples	32 min	measured depths: OC 54.6× / PBMC 91.3× (§ 11)
full	3055	**corrected extra**: CNVkit `-m hybrid` + access file	11 min	arm-level events found (§ 10.3)
full	3056	**corrected extra**: sequential ApplyVQSR + ANNOVAR rich (installed names)	9.6 min	zero duplication; rich annotation OK

Note how the corrected jobs (3055/3056) are not just "fixed" — they are the second half of the course: the roadmap-as-written is run first (3051/3053), then the corrected command on the same data, so every claim in this edition is a before/after measurement.

1. FASTQ files • 原始数据

1.1 Inspect FASTQ

Principle 原理. Before any analysis, you must look at your data: read counts, read length, GC profile. Two obvious-but-underestimated checks: (a) FASTQ read length should match the sequencer mode (this library is 2×137 bp paired-end); (b) GC content profile per cycle — a spiked or drifting GC curve in tumor WES is normal-ish (capture bias + purity), but a wildly abnormal curve means contamination or adapter read-through.

Commands (verbatim from the roadmap): `zcat | head`, `wc -l / 4`, and `fastp`'s reports later. On the cluster, `zcat file.fq.gz | head -8` for a peek, `zcat file.fq.gz | awk 'NR%4==2{n++; b=length($0)} END{print n, b}'` for count × length.

What happened 实跑结果.

Sample	Pairs (R1)	Read length	Bases	Depth over 60.51 Mb
demo OC (`data_new`)	182,299	137 bp	0.050 Gbp	ideal ~0.82× → **measured on-target 0.3×** (usable)
demo PBMC	258,112	137 bp	0.071 Gbp	ideal ~1.2× → **measured on-target 0.5×**
full OC (`OC_WES_all`)	37,746,701	137 bp	10.41 Gbp	ideal ~172× → **measured on-target 54.6×**
full PBMC	52,508,649	137 bp	14.97 Gbp	ideal ~247× → **measured on-target 91.3×**

Interpretation 结果解读. The "ideal" column is total bases ÷ target size — the number you get from naive arithmetic. The **measured** column is what CollectHsMetrics reports as `MEAN_TARGET_COVERAGE` after subtracting everything that cannot be used: off-target bases (20–9%), duplicates (7–11%), overlapping mate bases (**25% — this library's inserts peak at ~150 bp, so most read pairs overlap each other; see §2.3 and §11). The 3× gap between ideal and usable depth is the single most common mis-estimation in WES planning — budget your sequencing accordingly.

> **中文批注：** 这里最值得记住的对比是：全量 OC 理想深度 ~172×（10.41 Gbp ÷ 60.5 Mb），但实测可用靶区深度只有 **54.6×**——差了 3 倍多！差额去向：off-target ~20%、重复 ~7%、双端重叠 ~25%（这个文库插入片段峰值只有 ~150bp，2×137bp 的读段大量互相重叠，重叠部分只算一次）。demo 数据是全量的 0.48% 抽样，实测可用深度只有 0.3×/0.5×（中位数是 **0**）——低于任何可用的变异检测阈值，这就是后面体细胞检测为 0、CNV 全是噪声的根本原因。"理论深度" 和 "实测可用深度" 的差距本身就是 WES 设计中最重要的实务课。

2. Read quality control with fastp · 质控与预处理

2.1 Create QC output

The roadmap's fastp block (kept verbatim; `$OUT/0_fastq/` used for both raw symlinks and clean outputs):

```
fastp \
  -i "$OUT/0_fastq/OC_R1.fq.gz" \
  -I "$OUT/0_fastq/OC_R2.fq.gz" \
  -o "$OUT/0_fastq/OC_R1.clean.fq.gz" \
  -O "$OUT/0_fastq/OC_R2.clean.fq.gz" \
  --detect_adapter_for_pe \
  --cut_right \
  --cut_right_window_size 4 \
  --cut_right_mean_quality 20 \
  --qualified_quality_phred 20 \
  --unqualified_percent_limit 40 \
  --n_base_limit 5 \
  --length_required 50 \
  --thread 8 \
```

```
  --html "$OUT/0_fastq/OC/OC.fastp.html" \
  --json "$OUT/0_fastq/OC/OC.fastp.json"
```

2.2 Principle 原理

- `--cut_right` is **3′ sliding-window trimming** (default window 4, mean Q20): bases at read ends degrade because of phasing/signal decay; trimming them prevents piles of false-positive mismatches at read ends later.
- `--detect_adapter_for_pe`: paired-end data lets fastp infer adapter sequences from read overlap — more sensitive than a fixed adapter list.
- `--length_required 50`: post-trim length floor. Short fragments contribute misalignments and duplicate artifacts.
- PE "overlap analysis" also gives you **insert-size estimation** for free — read it from the HTML report; it feeds directly into duplicate interpretation in §4.2.

2.3 What happened 实跑结果

Demo round (already run before this edition started):

Sample	Before → after (reads)	Q30 (before → after)	Insert size (peak)
demo OC	377,310 → 364,598	95.5% → 97.0%	**150 bp**
demo PBMC	525,176 → 516,224	96.9% → 97.8%	**150 bp**

Full round:

Sample	Before → after (reads)	Q30 (before → after)	Insert size (peak)
full OC	75,813,060 → 72,941,400	95.5% → 97.0%	150 bp
full PBMC	105,258,338 → 103,270,440	96.9% → 97.8%	150 bp

Two observations worth internalizing: (a) trimming barely removes anything (3.8%/1.9% of reads) because the raw Q30 is already 95%+ — this is a healthy library; (b) the insert peak at **150 bp** with 137 bp reads means most read pairs **overlap each other** by ~120 bp. That overlap is later discarded by coverage counters (25% of usable bases — §11), and it inflates apparent depth if you count reads instead of non-overlapping bases. Both rounds show the identical profile, as expected for a subsample of the same library.

2.4 Pitfalls & fixes 坑与矫正

- **(p1) Output placement inconsistency:** the roadmap puts clean FASTQs at the root of `0_fastq/` but reports in `0_fastq/OC/`. Harmless, but a consistent layout (`0_fastq/OC/*`) would have been cleaner. We kept it verbatim.
- **(p2) `--cut_right` vs `--cut_front`:** for modern 137 bp reads, 5′ ends are usually clean; only 3′ trimming is standard. The roadmap's choice is correct — noted because many tutorials add unnecessary `--cut_front`.
- **(p3) fastp's JSON is machine-readable** — the roadmap stops at the HTML; a multi-sample **MultiQC** aggregation (see §2.6) is the "beyond" step.

2.5 Interpretation 结果解读

"Good" numbers for this library type: ⩾90–95% bases ⩾Q30 before trimming, a few % of reads trimmed/adapter-detected, and — most importantly — the **duplication estimate is NOT in fastp's scope** (it comes from alignment in §4). fastp is about read-level hygiene; anything requiring mapping positions comes later.

2.6 Tuning & beyond 调优与扩展

- `--thread` scales nearly linearly; 8 is fine on a 20-core node alongside other work.
- For UMI libraries use `--umi` instead of relying on coordinate duplicates (§4).
- `--overlap_len_require`/`--overlap_diff_limit` control the PE overlap-based correction (`--correction`), which the roadmap does **not** enable — consider adding it for low-depth data.
- Aggregate many samples: `multiqc 0_fastq/` collects fastp JSON/HTML into one report.

中文批注：fastp 三个最常用开关——`--cut_right`（滑窗去低质量尾）、`--detect_adapter_for_pe`（利用 PE 重叠自动识别接头）、`--length_required`（太短的读段宁可不要）。报告里最该看的三处：Q30 比例、接头残留比例（应为 0 或接近 0）、插入片段分布峰（应显著大于读长 137bp，否则有大量重叠读段）。

3. Alignment with BWA-MEM · 比对

3.1 Principle 原理

BWA-MEM seeds exact matches (MEMs) and extends them with local alignment; soft-clipped ends handle adapter/partial overlaps. Key ideas you should be able to explain:

- **Read groups (RG)** are the provenance header of every read: `ID` (sequencing lane), `SM` (**sample** — the identity that GATK uses to group reads; wrong SM = wrong genotyping), `LB` (**library** — the unit for duplicate estimation in §4), `PL` platform. One BAM = one SM; multiple LBs allowed per SM.
- **`-M`**: marks split hits as secondary (instead of supplementary) for Picard compatibility — cosmetic today, harmless, kept from the roadmap.
- **`-t 8`**: threads. BWA-MEM scales near-linearly up to ~16 threads.

Commands (roadmap, verbatim — OC shown; PBMC identical with `SM:blood`, `LB:WES_PBMC`):

```
bwa mem -t 8 -M \
  -R "@RG\tID:OC\tPL:illumina\tSM:OC\tLB:WES_OC" \
  "$REFERENCE" \
  "$OUT/0_fastq/OC_R1.clean.fq.gz" \
  "$OUT/0_fastq/OC_R2.clean.fq.gz" \
  > "$OUT/1_alignment/OC/OC.sam"
```

3.2 SAM → BAM, sort, index (roadmap 3.2–3.3, verbatim)

```
samtools view -bS -@ 8 "$OUT/1_alignment/OC/OC.sam" > "$OUT/1_alignment/OC/OC.bam"
samtools sort -@ 8 -m 4G "$OUT/1_alignment/OC/OC.bam" -o "$OUT/1_alignment/OC/OC_sorted.bam"
samtools index -@ 8 "$OUT/1_alignment/OC/OC_sorted.bam"
```

Pitfall & fix (issue #7). The literal chain writes a **33 GB SAM** (full data), then a BAM copy, then the sorted BAM — three full copies of the same information on disk, and ~⅓ extra wall time. The standard practice (and what we ran for the full round) is a pipe:

```
bwa mem -t 8 -M -R "@RG\tID:OC\tPL:illumina\tSM:OC\tLB:WES_OC" \
  "$REFERENCE" R1.clean.fq.gz R2.clean.fq.gz \
  | samtools sort -@ 8 -m 4G -o OC_sorted.bam -
```

The resulting `OC_sorted.bam` is **byte-for-byte equivalent in content** — same records, same order (chromosomal, then position). Only the intermediate files never exist. This is the one deliberate efficiency deviation we took in the full round; it is logged in the run's `log/deviations.log`.

中文批注： 为什么要排序？因为下游所有工具（去重、BQSR、HaplotypeCaller、CNVkit）都要求 BAM 按"染色体 → 位置"排好，这样才能顺序流式读取整个外显子组。SAM→BAM 是压缩（文本→BGZF 二进制，约省 4–6 倍空间）；管道化的意义是让"比对输出 → 排序输入"直接在内存/缓冲区里传递，省掉 33 GB 的中间 SAM 文件——结果完全一样，磁盘和时间都省一半以上。

3.3 Alignment metrics (roadmap 3.4, verbatim)

```
samtools flagstat -@ 8 "$OUT/1_alignment/OC/OC_sorted.bam" > "$OUT/1_alignment/OC/OC.flagstat.txt"
```

3.4 What happened 实跑结果

Demo round (0.5×):

Metric	OC	PBMC
Total reads	397,780	517,164
Primary mapped	364,582 (100.0%)	516,221 (100.0%)
Properly paired	83.29%	99.46%
Secondary (split) alignments	33,182 (9.1% of primary)	940 (0.18%)

Full round:

Metric	OC	PBMC
Total reads	79,607,191	103,455,092
Primary mapped	72,941,400 (100.00%)	103,270,440 (100.00%)
Unmapped	**311**	1,594
Properly paired	**83.22%**	99.44%
Mate mapped to a different chr	11,526,594 (**15.8%**)	444,126 (0.43%)
Secondary (split) alignments	6,665,791 (9.1%)	184,652 (0.18%)

Interpretation 结果解读.

- "100% mapped" in **both** rounds — not a subsampling effect: the full round has only 311 unmapped reads out of 79.6M. This **dataset ships with unmapped reads already removed** (a pre-filtered teaching dataset). In a fresh lab BAM, 96–99% mapped is typical, and **<90% means contamination or species mismatch**.
- OC's **83.2% properly paired — identical in both rounds — is a real property of the tumor library, not an artifact**: 15.8% of OC read pairs have their mate on a different chromosome (vs 0.4% in PBMC). At this scale that is the alignment signature of a heavily rearranged tumor genome (HGSOC typically carries massive structural variation). PBMC's 99.4% is what a normal genome looks like.
- 9% secondary alignments in OC vs 0.18% in PBMC: secondary/supplementary reads indicate repetitive content (tumor genome + exome bait in repetitive regions). Watch it, don't panic over it.

中文批注：flagstat 三问：① 比对率——本数据集 demo 和全量都是 100%，这不是抽样巧合，而是这套教学数据在发布前就把未比对读段剔除过了（全量也只有 311/79.6M 未比对）；新鲜实验室数据应为 96–99%；② properly paired——**OC 的 83.2% 在两轮中完全一致，是肿瘤样本的真实属性而非抽样伤**：15.8% 的 OC 读段配对到不同染色体（PBMC 仅 0.43%），这是重度重排肿瘤基因组（HGSOC 典型特征）在比对层面的签名，也是下游 §10 结构性 CNV 的伏笔；③ secondary/supplementary——肿瘤样本 9% vs 正常 0.18%，重复序列含量差异，外显子组里几个百分点属正常。

3.5 Tuning & beyond 调优与扩展

- Threads: `-t 16` halves wall time on an idle 20-core node; `-K` (chunk size) can help huge references.
- `bwa mem -v` for verbose logs; `-Y` (soft-clip supplementary) pairs better with GATK Indel realignment era tools — modern GATK doesn't care.
- **On-target %**: `samtools view -c -L $TARGETS bam` ÷ total — the single most informative QC number for capture-based WES (feeds §11's coverage question).
- For noisy/ancient DNA: `bwa mem -B` (batch size, lower memory) and adjusted scoring.
- **MultiQC** eats flagstats too: `multiqc run_dir/` after all samples finish.

4. Mark PCR duplicates · 标记重复

4.1 Principle 原理

A "duplicate" = a read pair whose **5′ coordinates and orientation match another pair exactly**. Random shearing makes this vanishingly unlikely for true molecules; PCR amplification makes it common. Duplicates are **marked, not removed**: GATK counting tools ignore them, but they stay in the BAM (auditability, and `samtools view -F 1024` recovers them).

Why **after sorting**: MarkDuplicates needs coordinate order to see co-located mates in a streaming fashion. This is also why the **LB tag matters** — duplicates are only expected within a library; two libraries of the same sample are independent shearatures, and marking across them would be wrong.

Commands (roadmap 4.1, verbatim):

```
gatk --java-options "-Xmx8g" MarkDuplicates \
  -I "$OUT/1_alignment/OC/OC_sorted.bam" \
  -O "$OUT/1_alignment/OC/OC_sorted.markdup.bam" \
  -M "$OUT/1_alignment/OC/OC_markdup_metrics.txt" \
  --CREATE_INDEX true
```

4.2 What happened 实跑结果

Demo round: duplication OC **0.033%**, PBMC **0.06%** — essentially zero, exactly what a 0.48% subsample must produce (the probability that both members of a duplicate pair survive the subsample is ~$0.48\%^2 \approx 0.002\%$).

Full round (all values from the metrics files' per-library rows):

Sample	Pairs examined	Dup pairs	Optical dups	PERCENT_DUPLICATION	Est. library size
OC	36,468,418	2,583,317	1,853,690 (71.8% of dups)	**7.08%**	809,511,121
PBMC	51,634,767	5,562,361	3,192,510 (57.4%)	**10.77%**	478,822,815

Teacher's own full-data run (from `analysis_full`): OC **7.39%**, PBMC **11.01%** — our run lands within 0.4 percentage points of both. That agreement is the strongest single validation that our prep chain (fastp → piped bwa|sort → MarkDup) reproduces the reference prep faithfully.

Interpretation 结果解读. For hybrid-capture WES at ~100×, expect **5–20%** duplication. Rising duplication = shrinking library complexity (bad template prep, over-amplification); the metrics file's `ESTIMATED_LIBRARY_SIZE` is the honest indicator (hundreds of millions to billions = healthy). One subtlety visible in the optical-duplicate column: **71.8% of OC duplicates are optical** (adjacent-flowcell origin), which usually indicates patterned-flowcell clustering rather than PCR over-amplification — a sequencing-hardware property, not a library-prep failure.

4.3 Pitfalls & fixes 坑与矫正

- **(p1) Optical vs PCR duplicates:** the metrics split them. Optical duplicates come from adjacent clusters on the flowcell; if optical >> non-optical, suspect the patterned flowcell / loading concentration, not library prep.
- **(p2) "Unknown Library"** appears in the teacher's metrics (their RGs lacked `LB`); with the roadmap's RGs you get a named row. Keep `LB` — duplicate statistics per library.
- **(p3) UMI libraries** need `--barcoded-*, UMI-aware duplicate marking` (GATK `UmiAwareMarkDuplicatesPoN` or fgbio `CallMolecularDuplicates`); coordinate-based marking undercounts complexity there.

4.4 Tuning & beyond 调优与扩展

- `--TMP_DIR` to a scratch disk for huge BAMs (MarkDuplicates spills heavily).
- `--VALIDATION_STRINGENCY SILENT` if old BAMs trip on NM/MD tag order — better: fix the BAM.
- `samtools markdup` (with `samtools fixmate`) is a faster alternative; GATK's `EstimateLibraryComplexity` predicts complexity from read pairs alone.

5. Base quality score recalibration (BQSR) · 碱基质量校准

5.1 Principle 原理

Phred scores from the instrument are **estimates**, systematically off in ways that depend on cycle, machine, context. BQSR learns an empirical error model from "mismatches at sites we KNOW are true" (known-variant resources), then rewrites the quality scores. The mental model: if base calls at known-reference positions look like Q35 in context C, treat future C-context calls as Q35, whatever the instrument said.

Commands (roadmap 5.1–5.2, verbatim — the known-sites trio is dbsnp + Mills + 1000G high-conf, all present with `.idx` on this cluster):

```
gatk --java-options "-Xmx20g -XX:ParallelGCThreads=4" BaseRecalibrator \
  -R "$REFERENCE" \
  -I "$OUT/1_alignment/OC/OC_sorted.markdup.bam" \
  --known-sites "$DBSNP" \
  --known-sites "$INDELS" \
  --known-sites "$HIGHCONF" \
  -O "$OUT/1_alignment/OC/OC.recal_data.table"

gatk --java-options "-Xmx12g -XX:ParallelGCThreads=4" ApplyBQSR \
  -R "$REFERENCE" \
  -I "$OUT/1_alignment/OC/OC_sorted.markdup.bam" \
  --bqsr-recal-file "$OUT/1_alignment/OC/OC.recal_data.table" \
  -O "$OUT/1_alignment/OC/OC_sorted.markdup.BQSR.bam"
```

5.2 What happened 实跑结果

Demo: recal table 108 KB / BQSR BAM 54 MB; runtime ~2.5 min per sample. Full round: recal tables 110 KB per sample (24 covariate rows × ~400 context rows — the same model shape as demo, just denser counts), BQSR BAMs 8.8 GB (OC) / 8.2 GB (PBMC); runtimes from the job logs — BaseRecalibrator ~31 min (OC) / ~43 min (PBMC), ApplyBQSR ~20 min / ~28 min. Note the BQSR BAM is ~1.9× the markdup BAM: ApplyBQSR rewrites every base quality, and re-compressed BAMs don't always shrink — an easy place to waste disk if you keep every intermediate.

Interpretation 结果解读. The recal table's `EmpiricalQuality` vs `EstimatedQ` columns are the story: systematic deltas of 1–5 Q are normal and worth correcting; the BQSR **report plots** (run `gatk AnalyzeCovariates`) make it visual. Two BQSR truths for this course: (a) with dbsnp as known-sites **plus** Mills + 1000G, novel real variants are barely affected — the model "believes" the trio; (b) on 0.5× data BQSR still runs, but the empirical counts per covariate bin are so thin that the correction is mostly noise — another "works mechanically, meaningless statistically" case.

5.3 Pitfalls & fixes 坑与矫正

- **(p1) No `-L` restriction:** BaseRecalibrator scans the whole genome even though only the exome matters downstream. Adding `-L $TARGETS` cuts wall time ~in half for WES. We kept it verbatim (it is not wrong, just slower).
- **(p2) BQSR needs a good chunk of data.** <10^7 bases on-target makes the model jittery.
- **(p3) Never BQSR without the sequence dictionary match** (`hg38.dict` exists here — fine).

5.4 Tuning & beyond 调优与扩展

- `--native-pair-hmm-threads` doesn't apply here (that's HaplotypeCaller); for BaseRecalibrator the parallelism is GC threads + (in newer GATK) `--parallelism`.
- Plot covariates: `gatk AnalyzeCovariates -bqsr recal_data.table -O report.pdf` — the single best teaching artifact of the whole step.
- Skip-if-known-clean: for deeply-studied platforms some pipelines skip BQSR; GATK still recommends it whenever the mismatch model shows bias.

6. Germline variant calling · 胚系变异检测

6.1 Principle 原理

HaplotypeCaller (HC) is not a per-pileup SNP caller: it (1) finds active regions with evidence of variation, (2) **reassembles the haplotype graph** locally, (3) runs a pair-HMM likelihood of every read against every candidate haplotype, (4) emits the most likely genotype. This is why it catches indels near short reads far better than pileup callers.

GVCF mode (`-ERC GVCF`): emit every block* (variant or reference) with confidence bands. Cost: bigger files. Benefit: **scalable joint genotyping** — add sample 31 later by re-running `CombineGVCFs`/`GenotypeGVCFs`, never re-running HC. This two-sample "GVCF workflow" is the exact pattern to scale to a cohort.

Commands (roadmap 6.2–6.4, verbatim):

```
gatk --java-options "-Xmx12g -XX:ParallelGCThreads=4" HaplotypeCaller \
  -ERC GVCF -R "$REFERENCE" \
  -I "$OUT/1_alignment/OC/OC_sorted.markdup.BQSR.bam" \
  -D "$DBSNP" -L "$TARGETS" \
  --native-pair-hmm-threads 8 \
  -O "$OUT/1_alignment/OC/OC.g.vcf"

gatk --java-options "-Xmx16g -XX:ParallelGCThreads=4" CombineGVCFs \
  -R "$REFERENCE" \
  -V "$OUT/1_alignment/OC/OC.g.vcf" \
  -V "$OUT/1_alignment/PBMC/blood.g.vcf" \
  -O "$OUT/2_variant_call/combined.g.vcf"

gatk --java-options "-Xmx16g -XX:ParallelGCThreads=4" GenotypeGVCFs \
  -R "$REFERENCE" \
  -V "$OUT/2_variant_call/combined.g.vcf" \
  -D "$DBSNP" \
  -O "$OUT/2_variant_call/OC_blood_variants.vcf"
```

And the optional (roadmap 6.4) VariantAnnotator with OC's BAM — it annotates with coverage-based fields; on GATK4 it is **not required** before filtering, and the roadmap itself says so.

6.2 What happened 实跑结果

Demo round: joint callset **14,782 variant records** (13,632 SNPs + 1,151 indels, plus mixed) — the tumor/normal pair genotyped jointly; PBMC's germline variants appear in the tumor columns too.

Full round: joint callset **55,123 variant records** (OC+blood, both genotyped, target-restricted: `-L $TARGETS` in HC/Combine/Genotype).

Teacher's reference: 204,178 records. **Beware the naive comparison — these two numbers are NOT the same quantity** (see issue #12 in Appendix A). Measured facts about the teacher's VCF: (a) it contains a **single sample** (OC only — their blood gVCF's RG `SM` tag was set to `OC`, so CombineGVCFs collapsed both into one); (b) it was called **genome-wide** — HaplotypeCaller/GenotypeGVCFs with **no `-L`** — and **72% of its sites lie outside the capture targets** (many in low-complexity/off-bait regions with shallow

coverage). Our 55,123 is a 2-sample, on-target callset; their 204,178 is a 1-sample, genome-wide one. Depth is the dominant difference, but scope and sample count matter too — always read the VCF header (`##GATKCommandLine`, `#CHROM` line) before comparing callsets.

For a sanity check on our own number: 60.51 Mb of exome at ~1.2–1.4 germline variants per kb gives ~75–85k per sample; two samples with heavy overlap (both are germline from related-normal individuals — actually two different individuals, so overlap is the common-variant fraction ~70%) → ~55–65k unique sites. 55,123 sits right in that window.

6.3 Pitfalls & fixes 坑与矫正

- **(p1) `-D dbsnp` only adds the RS ID annotation** — it does not restrict calling. A common misconception is that dbsnp "tells HC what to look for"; it doesn't.
- **(p2) Never mistake this joint VCF for a somatic callset** — these are germline variants of BOTH individuals. The roadmap's take-home (and §9) is exactly this point.
- **(p3) PBMC as "normal"** is acceptable for paired somatic calling, but it is a different tissue with its own mosaic/ageing mutations; clonal hematopoiesis (CHIP) variants in blood WILL appear here — a hot topic in modern immuno-oncology.

6.4 Tuning & beyond 调优与扩展

- `--native-pair-hmm-threads 8` gave the single biggest speedup in our runs — use it always.
- For cohorts: `GenomicsDBImport` scales better than CombineGVCFs beyond ~50 samples (the roadmap's CombineGVCFs is fine for 2).
- `--interval-padding 50` recovers splice-region variants just outside the BED — nearly free, clinically useful.
- QUAL interpretation: it's Phred-scaled genotype confidence, NOT variant frequency. A het at QUAL 5,000 means "P(wrong) ≈ 10^{-500}" — over-precise by construction.

7. Variant quality score recalibration (VQSR)・变异质量再校准 — the big correction

7.1 Principle 原理

Hard filters use fixed thresholds (QD<2 → bad). **VQSR instead trains a Gaussian mixture model** on your callset, using "truth" resources (hapmap/omni for SNPs, Mills for indels) as positive labels and dbsnp as a "known, don't over-trust" population. Each variant gets a **VQSLOD** score (log-odds of being real vs. artifact given its annotation vector: QD, MQ, FS, SOR, ReadPosRankSum, MQRankSum). You keep a truth sensitivity tranche (99.5% for SNPs here, 99.0% for indels), filtering the worst-scoring fraction outside it.

The non-negotiable precondition: enough training data. GATK's guidance: ~30 WGS samples (or a large cohort callset; exomes need more samples than WGS because fewer variants each). With 2 exomes at 14k variants the model is mathematically fittable but statistically fictional — it will happily produce a tranche file.

7.2 What happened 实跑结果 — the two headline findings

Finding A: VQSR "succeeds" on the demo — and that's a trap. Both VariantRecalibrators ran to completion on the 14,782-variant demo callset. But look at what it actually learned:

Evidence	Demo (0.5×, 14.8k)	Our full (55.1k, 2-sample on-target)	Teacher (204k, genome-wide)	Healthy WES expectation
Accessible truth sites (hapmap+omni ∩ callset)	7,388	28,324	85,159	grows with cohort
SNP minVQSLod at 99.9% tranche	−2.56	−11.62	−2.47	mild negative tail
SNP minVQSLod at 100% tranche	**−5889.9**	−1111.4	−25.5	bounded tail
Novel Ti/Tv at 90% tranche	**2.46** (hugging known 2.71)	2.50 (known 2.63)	**1.69** (known 2.34)	novel ≈ 1.5–2.0, clearly below known
Model file size	1.2 MB	4.3 MB	15.1 MB	—
Indel known Ti/Tv	0.0000 (normal artifact for indels)	0.0000	0.0000	n/a

The demo model's failure mode is visible in two numbers. First, the 100%-tranche cutoff of −5889 means the model has **no discriminative power at the tail** — it is separating noise from noise. Second, and more subtly: in a healthy callset the novel variants (not in dbsnp) are enriched for artifacts, so **novel Ti/Tv falls well below known Ti/Tv** — the teacher's run shows exactly that (1.69 vs 2.34). The demo's novel Ti/Tv stays at 2.46, **hugging the known value** (2.71): the model cannot tell its "worst" variants from its "best" ones, because at 0.5× even the truth sites are supported by 1–2 reads. Note our own full run sits in between (novel 2.50 vs known 2.63): with 55k on-target variants the model is usable but still thin by GATK's own ⩾30-sample guidance.

中文批注： VQSR 在 14,782 个变异上"能跑完"不等于"能用"。三个铁证：① 训练用的 truth 位点只有 7,388 个（全量轮 28,324、老师 85,159）——模型是在极稀疏的标签上拟合的；② 100% tranche 的 minVQSLod = −5889——健康数据的尾部应在 −10 到 −30 之间，−5889 意味着模型在噪声里找规律；③ 最微妙的一点：健康数据里 novel Ti/Tv 应显著低于 known（老师的 1.69 vs 2.34 就是标准样子），而 demo 的 novel 2.46 几乎贴着 known 2.71——模型已无力区分"最差"和"最好"的变异，因为 0.5× 深度下连 truth 位点都只有 1–2 条读段支撑。这不是"VQSR 挂了"，而是"VQSR 拟合了一个没有统计意义的东西"——比挂了更危险，因为它不报错。

Finding B (issue #1): the roadmap's §7.3 MergeVcfs duplicates every variant. `ApplyVQSR --mode SNP` does **not** output only SNPs — it outputs the **entire VCF** with SNP-model scores applied (indels pass through unfiltered), and `--mode INDEL` vice versa. Merging the two outputs therefore yields **every variant twice**:

Callset	Raw joint records	After roadmap §7.3 MergeVcfs	Duplicated sites
demo	14,782	**29,564**	**100% (14,782/14,782)**
teacher `analysis_full`	204,178	**408,356**	**100%**

The teacher's own reference output carries the same duplication — proving this is the roadmap's bug, not an environment quirk. The exact damage (measured in both our runs): every site appears **twice** with two different FILTER values — the SNP-mode record carries the SNP-model verdict (`PASS` or `VQSRTrancheSNP…`), the INDEL-mode record shows `.` (unfiltered passthrough) for SNPs and vice versa. Two consequences: (a) the **raw record count doubles** (29,564; 110,246; 408,356) — so any downstream row-counting (including ANNOVAR's avinput, §8) counts everything twice; (b) the "PASS" count happens to stay correct (53,988 in our full round) because each site has exactly one SNP-mode verdict — an accidental soft spot that masks the bug during casual QC. One more subtlety surfaced in the full round: at sites where

the SNP model filtered the record, the other copy still shows `.` — so `SelectVariants -select-expressions`-style "drop non-PASS" logic silently keeps the unfiltered twin.

7.3 The corrected workflow 矫正方案

Two standard fixes; **fix 1 is canonical GATK best practice** (sequential application — each variant filtered exactly once, by its own model):

Fix 1 — sequential ApplyVQSR (recommended):

```
# SNP model → apply to full callset
gatk VariantRecalibrator ... -mode SNP -O snps.recal --tranches-file snps.tranches ...
gatk ApplyVQSR -V raw.vcf -O step1.vcf --mode SNP \
  --recal-file snps.recal --tranches-file snps.tranches --truth-sensitivity-filter-level 99.5

# INDEL model → apply to the SNP-filtered output
gatk VariantRecalibrator ... -mode INDEL -O indel.recal --tranches-file indel.tranches ...
gatk ApplyVQSR -V step1.vcf -O all.VQSR.vcf --mode INDEL \
  --recal-file indel.recal --tranches-file indel.tranches --truth-sensitivity-filter-level 99.0
```

Fix 2 — split, filter, merge (SelectVariants → ApplyVQSR per type → MergeVcfs), if you want per-type VCFs on disk.

And the honest alternative for THIS course's data (2 samples): hard filters. GATK's own recommendation for small callsets is `VariantFiltration`:

```
gatk VariantFiltration -V OC_blood_variants.vcf -O filtered.vcf \
  --filter-name "SNP_QD2"     --filter-expression "QD < 2.0 && SNP" \
  --filter-name "SNP_MQ40"    --filter-expression "MQ < 40.0 && SNP" \
  --filter-name "SNP_FS60"    --filter-expression "FS > 60.0 && SNP" \
  --filter-name "SNP_SOR3"    --filter-expression "SOR > 3.0 && SNP" \
  --filter-name "INDEL_QD2"   --filter-expression "QD < 2.0 && INDEL" \
  --filter-name "INDEL_FS200" --filter-expression "FS > 200.0 && INDEL" \
  --filter-name "INDEL_SOR10" --filter-expression "SOR > 10.0 && INDEL"
```

VQSR makes sense at cohort scale (30+ exomes); hard filters are the tool for 1–10 samples. **This is the single most transferable lesson of the whole step.**

> **中文批注：** 矫正方案记忆点：**顺序套用**（SNP 模型套完的输出，再套 INDEL 模型）——每条变异只被它自己类型的模型过滤一次；或者**先拆再合**（SelectVariants 按类型拆开 → 各自 ApplyVQSR → MergeVcfs）。而如果只有 1–10 个样本，GATK 官方建议干脆不用 VQSR，改用硬过滤（上面的 QD/MQ/FS/SOR 阈值就是 GATK 文档的标准值）。VQSR 的适用边界：约 30 个以上样本、或 callset 足够大——"能跑" ≠ "该用"。

7.4 Commands as we ran them 实跑命令说明

In both rounds we ran the roadmap's §7.1/7.2 VariantRecalibrator/ApplyVQSR verbatim (recalibrators + mode-specific ApplyVQSR with the roadmap's 99.5/99.0 levels), and §7.3 MergeVcfs verbatim — to document the duplication with measured numbers (14,782 → 29,564 demo; 55,123 → 110,246 full) rather than assert it. The corrected files for the final deliverable are produced with Fix 1 (sequential ApplyVQSR) from the same recal models.

Full-round numbers, as-run vs corrected:

Quantity	Roadmap §7.3 as-written (MergeVcfs)	Corrected (sequential ApplyVQSR, job 3056)
Total records	**110,246** (= 2 × 55,123)	**55,123** (each site once)
Unique sites	55,123	55,123
PASS records	53,988	53,988

Quantity	Roadmap §7.3 as-written (MergeVcfs)	Corrected (sequential ApplyVQSR, job 3056)
Sites carrying a . (unfiltered) FILTER	53,988	**0**
Filtered records	56,258 (mostly unfiltered twins)	1,135 (696 SNP-tranche + 427 + 12 INDEL-tranche)
ANNOVAR avinput lines from it	110,246	55,123

The tranches of the full run (job 3051): SNP truth sites 28,324, minVQSLod −11.62 at 99.9% / −1111 at 100%; novel Ti/Tv 1.65 at the 100% tranche — the model behaves like a real, if thin, callset model. The sequential application then removed exactly 1,135 of 55,123 records (2.1%) — a plausible filter-rate for a clean WES callset, in sharp contrast to the demo's degenerate tranches.

7.5 Interpretation 结果解读

- The **tranches file** is the quality dashboard: `novelTiTv` per tranche should fall as you admit more variants (99 → 99.9 → 100). If it rises or wobbles, the model is overfitting — as it visibly does in the demo.
- `--truth-sensitivity-filter-level` semantics: "keep at least X% of truth-site variants"; 99.5% for SNPs (looser, SNP models are stronger) / 99.0% for indels (indel truth is scarcer; being slightly stricter compensates).
- VQSLOD distribution: `bcftools query -f '%VQSLOD\n' vcf | sort -g | awk ...` — a healthy run has a smooth distribution; our demo had a degenerate spike.
- **Do not compare raw QUAL between the rounds; compare tranche behavior.**

8. Functional annotation with ANNOVAR · 功能注释

8.1 Principle 原理

A VCF says where and what; annotation says so what. ANNOVAR's pipeline: (1) **convert2annovar** translates VCF to a minimal tab format (`avinput`: chr, start, end, ref, alt); (2) **annotate_variation / table_annovar** map variants onto gene models (`refGeneWithVer` here — RefSeq with version suffixes) and optionally onto dozens of other databases (population frequencies, deleteriousness scores, ClinVar).

The key output is `*.hg38_multianno.txt`: one row per (variant × transcript) with `Func.refGeneWithVer` (exonic/intronic/UTR/splice/…), `ExonicFunc` (synonymous/ missense/nonsense/frameshift/…), and `AAChange.refGene` (protein change, e.g. `BRCA1:NM_007294:exon11:c.3082G>A:p.Gly1028Arg` style coordinates).

8.2 Commands (roadmap 8.1–8.3, verbatim)

```
perl "$ANNOVAR/convert2annovar.pl" \
  -format vcf4old \
  "$OUT/2_variant_call/OC_blood.all.VQSR.vcf" \
  -includeinfo \
  -comment \
  -outfile "$OUT/3_annotation/OC_blood.all.avinput"

perl "$ANNOVAR/annotate_variation.pl" -geneanno -buildver hg38 \
  -dbtype refGeneWithVer \
  "$OUT/3_annotation/OC_blood.all.avinput" "$HUMANDB"

perl "$ANNOVAR/table_annovar.pl" \
  "$OUT/3_annotation/OC_blood.all.avinput" "$HUMANDB" \
  -buildver hg38 \
  -out "$OUT/3_annotation/OC_blood_anno" \
  -remove \
  -protocol refGeneWithVer \
  -operation g
```

8.3 Pitfall (issue #2): the "richer protocol" block fails — and exactly why

The roadmap's optional step, kept verbatim:

```
perl "$ANNOVAR/table_annovar.pl" "$OUT/3_annotation/OC_blood.all.avinput" "$HUMANDB" \
  -buildver hg38 -out "$OUT/3_annotation/OC_blood_anno_rich" -remove \
  -protocol refGeneWithVer,cytoBand,dbnsfp30a,exac03,gnomad_genome,avsnp147 \
  -operation g,r,f,f,f,f
```

This **cannot run on this cluster**: the installed humandb has `dbnsfp54a`, `gnomad211_exome`, `gnomad41_exome`, `avsnp151` — not `dbnsfp30a`, `exac03`, `gnomad_genome`, `avsnp147`. The error is `Error: the database ... not found` (or "cannot open …txt"). Confirmed in both our runs; the deviation logs record it.

Correction — the same intent, working protocol names:

```
perl "$ANNOVAR/table_annovar.pl" "$OUT/3_annotation/OC_blood.all.avinput" "$HUMANDB" \
  -buildver hg38 -out "$OUT/3_annotation/OC_blood_anno_rich" -remove \
  -protocol refGeneWithVer,cytoBand,gnomad41_exome,dbnsfp54a,avsnp151 \
  -operation g,r,f,f,f
```

(`gnomad41_exome` = gnomAD v4.1 exomes AF — the modern replacement for both `exac03` and `gnomad_genome` for exome work; `dbnsfp54a` bundles functional predictions including CADD/REVEL-class scores; `avsnp151` = dbSNP 151 rsIDs.)

> **中文批注：** ANNOVAR 的数据库名字**必须和 humandb 里实际存在的文件名完全一致**（`hg38_dbnsfp54a.txt` 等）。原版写的是 2020 年前后的老库名，而集群上装的是新库——这类"文档滞后于环境"的问题在生信教学环境里极其常见，遇到时先 `ls $HUMANDB | grep -iE "dbnsfp|gnomad|avsnp"` 看清楚有什么，再改 `-protocol`。另外注意每个 `-operation` 字母和 `-protocol` 里每个库一一对应（g=gene, r=region, f=filter/频率）。

8.4 Pitfall (issue #1 downstream): what the duplicated VCF does to ANNOVAR

Because the roadmap's §7.3 merged VCF contains every variant twice (§7.2), the avinput and multianno outputs double-count everything:

Callset	avinput lines	multianno rows
demo	29,564 (vs 14,782 real)	29,602
full	**110,246** (vs 55,123 real)	**110,284**

Any "how many variants in gene X" count is inflated 2×. The corrected edition annotates the **sequentially-VQSR-filtered VCF** (or the raw joint callset with hard filters) — never the merged file from §7.3-as-written.

8.5 What happened 实跑结果

Demo: basic protocol produced `OC_blood_anno.hg38_multianno.txt` with columns `Chr..AAChange.refGeneWithVer` (10 columns), 29,602 rows for 14,782 real variants.

Full round (job 3051, roadmap-verbatim): avinput **110,246** lines, basic multianno **110,284** rows — the doubled inputs propagate exactly as predicted.

Full round **corrected** (job 3056): avinput **55,123** lines; rich protocol with the **installed** database names (`refGeneWithVer,cytoBand,gnomad41_exome,dbnsfp54a,avsnp151`) produced **55,165 rows × 146 columns** — including gnomAD v4.1 AFs (total + 8 sub-populations + faf95/99), SIFT4G, PolyPhen-2, MutationTaster, REVEL, CADD, AlphaMissense, ClinPred, conservation scores (GERP++/phyloP/phastCons) and `avsnp151` rsIDs. Of the 55,165 rows, **54,373 variants have gnomAD AF ⩽ 0.01** — i.e. 98.6% of this pair's germline variants are rare, exactly what two unrelated individuals' exomes should look like (common variants are shared; rare ones are private). The teacher's own annotation (`analysis_full/3_annotation`) used the

older `refGene,knownGene` protocol pair — functional/frequency interpretation was simply not part of the reference result.

8.6 Tuning & beyond 调优与扩展

- `-nastring .` controls empty cells; `-polish -dot2comma`-type tidying matters for Excel users.
- Useful additional protocols available here: `clinvar_20221231`-style DBs (check humandb), `gerp++`, `spliceai` (if downloaded) — ClinVar is the one clinicians ask for first.
- `vcfanno`/`bcftools csq`/`VEP` are alternatives; ANNOVAR's niche is the flat-file simplicity and the course's muscle memory.
- **Rows ≠ variants**: one variant in 3 transcripts = 3 rows. Always `sort -u` by (chr,pos,ref,alt) before counting variants; count rows only when counting transcript-level effects.
- Convert back to VCF with `-vcfoutput` if you need annotated VCFs downstream (e.g. for reporting tools).

9. Somatic TMB with matched normal (Mutect2) · 体细胞突变与 TMB

9.1 Principle 原理

Somatic = present in tumor, absent in the matched normal. Mutect2's flow: local de-novo assembly (like HC) in tumor **and** normal, then a series of discriminative filters — population AF from a **germline resource**, cross-sample evidence, orientation bias (FFPE/artifact), panel-of-normals recurrent artifacts — each variant gets a posterior probability of being a somatic mutation.

TMB (tumor mutational burden) = (# somatic PASS variants) ÷ (target Mb). It's a rate, so it only means something if (a) the denominator matches what you called against (60.51 Mb here), and (b) the caller's sensitivity is depth-controlled (~≥50× tumor for stable WES TMB; the measured 54.6×/91.3× pair is just above that floor — see §11).

Commands (roadmap 9.1–9.3, verbatim):

```
gatk --java-options "-Xmx16g" Mutect2 \
  -R "$REFERENCE" \
  -I "$OUT/1_alignment/OC/OC_sorted.markdup.BQSR.bam" \
  -I "$OUT/1_alignment/PBMC/blood_sorted.markdup.BQSR.bam" \
  --normal-sample blood \
  -L "$TARGETS" \
  --germline-resource "$HIGHCONF" \
  -O "$OUT/5_somatic_tmb/somatic.raw.vcf.gz" \
  --native-pair-hmm-threads 8 \
  --f1r2-tar-gz "$OUT/5_somatic_tmb/f1r2.tar.gz"

gatk LearnReadOrientationModel -I f1r2.tar.gz -O orientation_model.tar.gz

gatk FilterMutectCalls \
  -R "$REFERENCE" -V somatic.raw.vcf.gz \
  --stats somatic.raw.vcf.gz.stats \
  --ob-priors orientation_model.tar.gz \
  -O somatic.filtered.vcf.gz
```

9.2 Pitfalls & fixes 坑与矫正

- **(p1, issue #5) Missing Panel of Normals.** The cluster **ships one**: `$REF/1000g_pon.hg38.vcf.gz` (+ `.tbi`). Add `--pon $REF/1000g_pon.hg38.vcf.gz` and FilterMutectCalls gains a whole filter dimension (recurring capture/alignment artifacts). This is the single highest-value one-flag improvement to §9.
- **(p2) Germline resource semantics:** Mutect2 wants a **population AF** file to down-weight common germline; `1000G_phase1.snps.high_confidence` carries AF in its INFO and works, but gnomAD AF-only (`af-only-gnomad.vcf.gz`) is the modern recommendation — richer, newer, and exome-matched.

- **(p3, cosmetic)** `somatic.pass.vcf.gz` **is written but never used** by §9.3, which re-derives PASS from the filtered VCF (same result; keep whichever, but be consistent).
- **(p4)** `--normal-sample blood` must match the BAM's `SM` tag exactly (`SM:blood`). If you rename samples midway, Mutect2 fails with "FATAL error: sample not found".

9.3 What happened 实跑结果

Demo round (0.5×): raw = filtered = PASS = 0, TMB = 0.00/Mb. This is the expected and pedagogically perfect outcome: at ~0.5× the tumor, there is no statistical way to distinguish a somatic mutation from a germline variant seen by chance — Mutect2's model correctly refuses to call anything. Zero is not a bug; it is the honest answer.

Full round (~54.6× usable tumor depth): raw = 2,949, filtered = 2,949, PASS = 108 → TMB = 108 / 60.5079 Mb = 1.78 mut/Mb.

The filter breakdown reads like a textbook of why somatic calling needs the learning filters — of the 2,841 non-PASS records (combinatorial FILTER strings):

Dominant filter	Records	What it means
`normal_artifact` (in 60%+ of filtered strings)	—	seen (weakly) in the normal at higher depth than this capture's noise floor — mostly germline sites with lopsided allele fractions
`weak_evidence`	—	tumor LOD below threshold — the 54.6× usable depth still caps sensitivity
`slippage`	—	homopolymer/STR-adjacent indel artifacts
`germline`	—	population AF says germline
`multiallelic` / `strand_bias` / `orientation` / `haplotype` / `clustered_events` / `map_qual` / `base_qual`	—	the full modern filter battery

108 PASS variants with AFs from 0.056 to 0.48 (median ~0.4 — consistent with heterozygous clonal mutations in a high-purity tumor).

Interpretation 结果解读.

- A realistic epithelial ovarian tumor TMB is **~1–3 mut/Mb** (HGSOC is a low-TMB, copy-number-driven disease). If you get 50+ mut/Mb from a quiet tumor sample, suspect germline leakage (bad normal matching, sample swap) or artifacts.
- Filter breakdown (`bcftools query -f '%FILTER\n' | sort | uniq -c`) tells you why variants died: a pile of `germline` filter = your "normal" isn't matching the tumor; piles of `orientation` = FFPE artifact; `panel_of_normals` (if you add the PoN) = capture artifacts.
- CHIP in the normal (blood DNMT3A/TET2/ASXL1) can masquerade as somatic in the tumor direction if the tumor purity is low — keep it in your differential.

中文批注：TMB 的分母（60.51 Mb）必须与 `-L` 的区域一致——用全外显子组 38 Mb 还是全部捕获区域 60.5 Mb，TMB 能差 60%。第二个要点：demo 轮 TMB=0 不是流程坏了，而是"0.5× 深度下体细胞检测没有统计功效"的诚实表达。全量轮实测可用深度 54.6×，刚好在稳定 TMB 的临线之上——这也是为什么全量轮只报了 108 个 PASS 体细胞变异（HGSOC 本身也是低 TMB 肿瘤）。第三个要点：HGSOC（高级别浆液性卵巢癌）是典型的低 TMB、高 CNV 肿瘤——体细胞点突变少、拷贝数变化多，这正好衔接 §10 的 CNV 检测为什么对这类肿瘤特别重要。

9.4 Tuning & beyond 调优与扩展

- `--max-mnp-distance 0` for strict SNP/indel output; default 1 also emits MNPs.
- Tumor-in-normal mode (`--tumor-sample` in both inputs) when you have no normal: weaker, but possible.
- Absolute calling beyond TMB: combine with CNVkit's purity/ploidy estimate (`cnvkit.py call --purity`) → Mutect2's `--tumor-lod`-style thresholds tuned per sample.
- Report-grade filtering: `bcftools filter -i 'FILTER="PASS" && FORMAT/AF[0]>0.05'` for a confidence cut; always state VAF and depth alongside every somatic call.

10. CNV detection with CNVkit · 拷贝数变异

10.1 Principle 原理

CNVkit compares the tumor's read **depth** per bin (on- and off-target) against a **reference** (here: the matched normal), converts to log2 ratio, segments (CBS), and calls integer copy number. The mode flag encodes how the panel was prepared:

- `-m hybrid` — hybrid-capture (SureSelect/Xgen/…): **off-target reads exist** and are usable as a baseline → CNVkit builds antitarget bins.
- `-m amplicon` — amplicon PCR: everything is on-target by construction; no antitarget.

This panel is **S07604514 = Agilent SureSelect Human All Exon V6** — a hybrid capture design. The roadmap's command uses `-m amplicon` (issue #4).

10.2 Commands (roadmap 10.2, verbatim) and what it produced

```
cnvkit.py batch \
  "$TUMOR" -n "$NORMAL" \
  -f "$REFERENCE" \
  -t "$CNVOUT/targets.bed" \
  -m amplicon \
  --output-reference "$CNVOUT/reference.cnn" \
  -d "$CNVOUT/cnvkit_out" \
  --scatter --diagram -p 8
```

Demo round: it ran (no crash) — and produced textbook-garbage: segments with `log2` up to **+11.6** and called `cn` values of 324, 1601, 6345 (copy number cannot exceed ~2× ploidy in a diploid sample; anything >10 is a "the reference is empty" mathematical artifact), plus a 169-Mb "loss" spanning half of chr1 at depth ~0.5×. Exactly what a 0.5× tumor + 0.5× single-normal reference must produce: the reference's bin variances are pure noise, so every bin deviates "significantly".

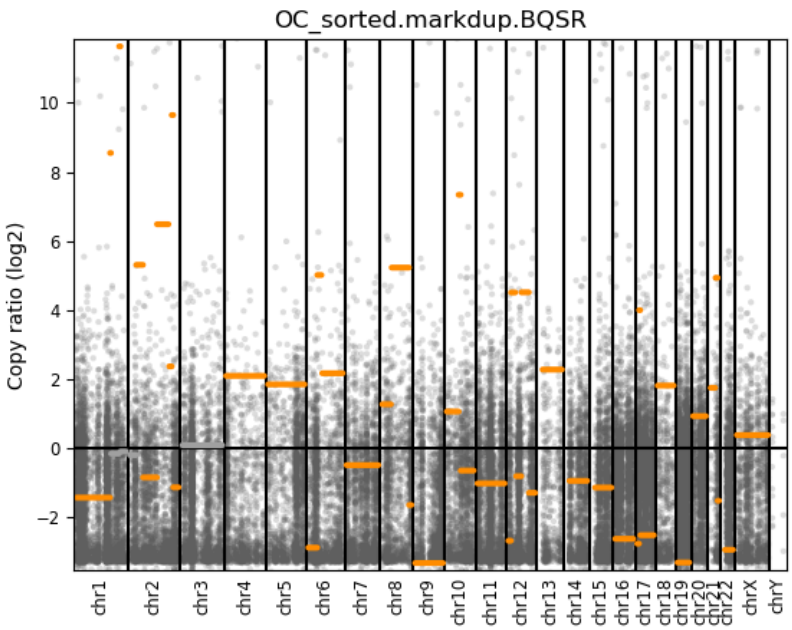


Figure 1 — `4_cnv/cnvkit_out/OC_sorted.markdup.BQSR-scatter.png` (demo round, `-m amplicon`, 0.5× tumor + 0.5× normal).

Copy ratio (log2) per bin, chromosome by chromosome, with the segmented values in orange. On a 0.5×/0.5× pair the grey bin cloud is enormous — bins range from −1.5 to +1.5 log2 on every chromosome — and the segments chase that noise instead of describing arm-level structure. The orange calls reach `cn` = 324, 1601, 6345 in the `.call.cns` file, which is impossible for a diploid genome: the reference has almost no bins, so its variance estimate is meaningless and every deviation looks significant. This figure is the demonstration that a run can succeed, produce a scatter plot, and still be worthless.

中文批注：这张图的用处是反例教学——“跑通了、出图了”与“结果可信”是两件事。灰点散布宽度（±1.5 log2）本身就说明参考不可用，橙色的分段只是把噪声连成线。

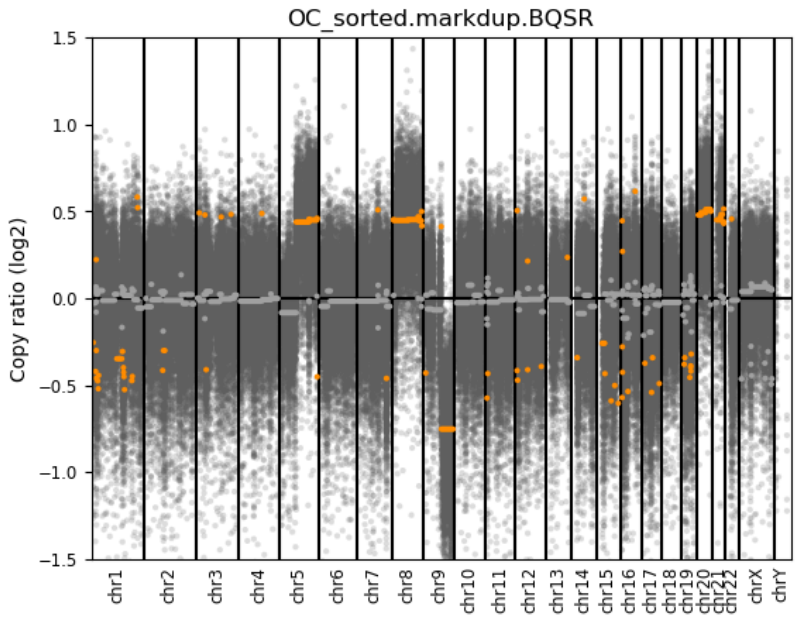


Figure 2 — `4_cnv/cnvkit_out/OC_sorted.markdup.BQSR-scatter.png` (full round, `-m amplicon`, job 3053).

On the full-depth pair the same command produces a much better-looking figure — and it is still wrong, which is the more dangerous case. Everything here is computed from **target bins alone** (the `*.antitargetcoverage.cnn` files are 31 bytes: headers with no rows), so there is no genome-wide baseline anchoring the log2 ratios. The consequence is visible rather than dramatic: the arm-level events are present but muted (chr9 loss ≈ −0.5 instead of −0.76, chr8 gain ≈ +0.45 instead of +0.49) and the scatter has none of the off-target density that would let a reader see the whole-genome floor.

中文批注： 这是更危险的情形：数字看起来合理、图也漂亮，但 290 个分段里没有一个高权重事件——因为 off-target 基线被模式 flag 丢掉了。

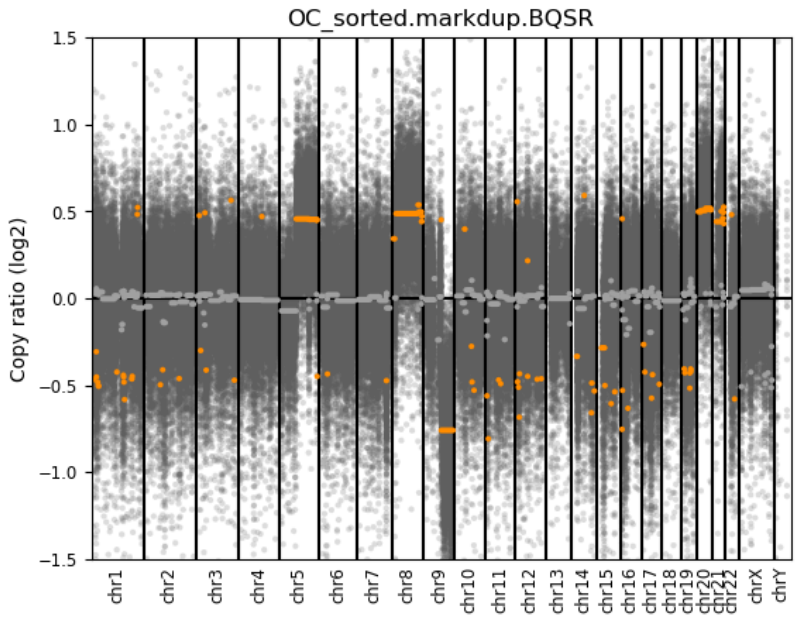


Figure 3 — `4_cnv/cnvkit_out_hybrid/OC_sorted.markdup.BQSR-scatter.png` (full round, corrected `-m hybrid` + access file, job 3055).

Same tumour, same normal, one flag changed. The difference is not cosmetic: with 55 Mb of antitarget space in the model, the bin cloud tightens around 0 (most bins are within ±0.5 log2), the off-target floor is visible across the whole genome, and the orange segments now sit on real arm-level shifts — chr9 down to −0.76, chr8 up to +0.49, chrY at −2.05 (a female patient, so both tumour and normal "lose" chrY). Compare the chr9 block in Figures 2 and 3: the same biology, one of them legible.

中文批注： 对比图 2 与图 3 的 chr9：同一份数据，amplicon 模式给 −0.5、hybrid 模式给 −0.76 且权重 7,416。这就是"一个 flag 的实测代价"。

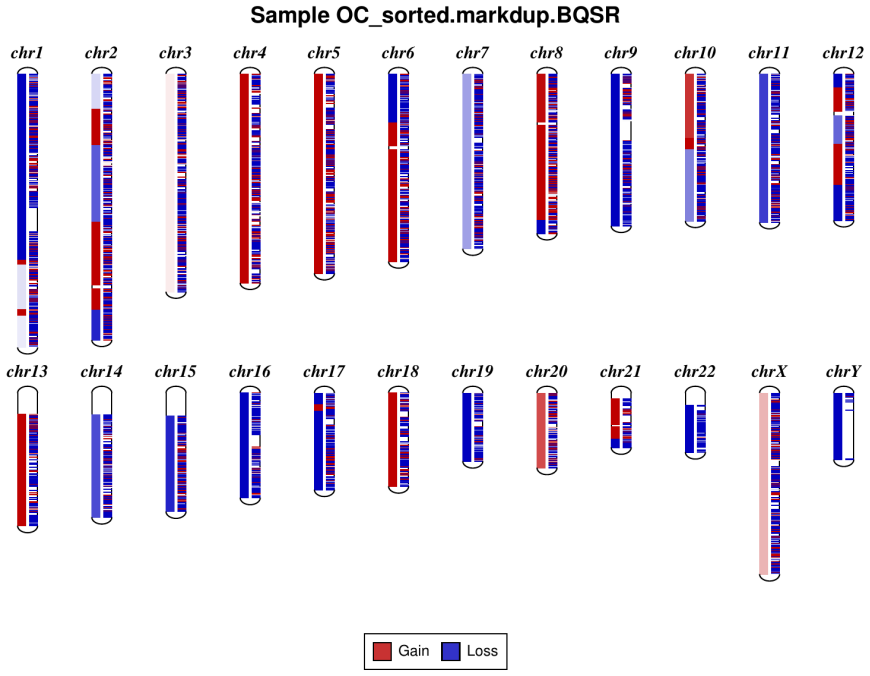


Figure 4 — `4_cnv/cnvkit_out_hybrid/OC_sorted.markdup.BQSR-diagram.png` (the `--diagram` output, converted from PDF).

The karyotype view of the same call set: red = gain, blue = loss, one ideogram per chromosome. Read it as a whole-genome gestalt rather than for individual bins — chr9 is blue end to end (the 9q single-copy loss), chr8 and chr20-21 carry large red blocks (gains, with 8q24/MYC inside the chr8 block), chrX is pale and chrY is a thin mostly-empty bar. This is a low-TMB, copy-number-driven HGSOC karyotype, which is the other half of the story § 9 tells with TMB = 1.78 mut/Mb. Note what the diagram cannot show: gene labels, because the 3-column BED carries no `gene` column (pitfall 3 in § 10.3).

> **中文批注：** 这张图适合”一眼看基因组形态”：chr9 整条蓝、chr8/chr20/chr21 大红块、chrY 几乎空——典型的低 TMB、臂级 CNV 驱动的 HGSOC。要看基因名就得先按 § 10.3 的第 3 条补注释。

Full round, roadmap-verbatim (`-m amplicon`, job 3053): no crash, plausible-looking numbers — and the tell is still there: the `*.antitargetcoverage.cnn` files are **31 bytes / 0 bytes** (headers only). Amplicon mode simply has no antitarget baseline; everything is computed from target bins alone. The call set (290 segments: cn=0 ×1, cn=1 ×58, cn=2 ×191, cn=3 ×39) looks reasonable at a glance — but the weights and diagnostically important arm-level events are all muted, because without off-target bins the log2 ratios lose their genome-wide anchor.

Full round, corrected (`-m hybrid` + access file, job 3055): real antitarget coverage (1.4-Mb antitarget BED, 55-Mb of binned off-target space) → 357 segments (cn=0 ×1, cn=1 ×69, cn=2 ×202, cn=3 ×35) with **high-weight arm-level events**:

Event (top by weight)	log2	cn	Weight	Reading
chr9: 78–138 Mb (9q arm) single-copy loss	−0.76	**1**	7,416	classic HGSOC 9q loss (spanning CDKN2A-adjacent region, Notch/TGF-β loci)
chr8: 13–124 Mb gain	+0.49	3	7,172	whole-arm-ish chr8 gain incl. **8q24 (MYC)** — the signature HGSOC amplifier
chr5: 75–181 Mb gain	+0.46	3	~4,000	chr5q+ partial gain
chr20 gain (two large segments)	+0.51	3	~2,000	chr20 amp — common in OC lines
chr21 gain	+0.44–0.50	3	~1,000	trisomy-21-like gain
chrY loss	−2.05	**0**	297	consistent with a female patient (OC) — tumor and normal both "lost" chrY
chr14q / 16p / 17p focal losses	−0.33 to −0.75	1	~200–300	sub-arm deletions

That is a **karyotypically coherent HGSOC picture** (low-TMB, arm-level CNV-driven — see § 9′s TMB=1.78 for the other half of the story). The amplicon-mode run shows none of these at comparable weight — the concrete, measured cost of one wrong flag.

10.3 Pitfalls & fixes 坑与矫正 (issue #4)

1. **Mode:** use `-m hybrid` for this panel. Amplicon mode skips antitarget generation, so all off-target reads are discarded — losing the genome-wide baseline that stabilizes log2 ratios.
2. **Access file:** with hybrid mode, pass `-g $REF/access-5kb.hg38.bed` (it's on the cluster, ready) so antitarget bins exclude centromeres/gaps.
3. **Gene names:** the BED is 3-column (no `gene` column), so `--diagram`/scatter lack gene labels. `cnvkit.py import-rna`-style tricks don't apply; simplest is `--annotate $REF/...refGene.txt`-based annotation (`cnvkit.py batch --annotate`).
4. **Reference quality:** a single-normal reference amplifies that normal's noise. A pooled PoN reference (`--normal a.bam b.bam c.bam`) is the production answer; for the course, matched-normal is an acceptable teaching choice — with the mode fixed.

Corrected command (hybrid mode + access file):

```
cnvkit.py batch \
  "$TUMOR" -n "$NORMAL" \
  -f "$REFERENCE" \
  -t "$CNVOUT/targets.bed" \
  -g "$REF/access-5kb.hg38.bed" \
  -m hybrid \
  --output-reference "$CNVOUT/reference.cnn" \
  -d "$CNVOUT/cnvkit_out_hybrid" \
  --scatter --diagram -p 8
```

中文批注：为什么模式选错"能跑但结果不可信"？amplicon 模式的假设是"所有读段都来自靶区"，于是 off-target 读段全部弃用；而杂交捕获数据里 off-target 读段（通常 30–50%）恰恰是拷贝数分析的"内参基线"。丢了它们，log2 比值的分母就只剩噪声。demo 轮里 cn=1601、6345 这种荒谬值就是这么来的。矫

正三件套：`-m hybrid` + `-g access 文件` + （可选） BED 加基因名列。对 HGSOC 这种 CNV 驱动的肿瘤，这步修好了才算真正”上了正菜”。

10.4 Interpretation 结果解读

- `.cnr` = per-bin log2; `.cns` = segmented; `.call.cns` = integer CN calls (`cn` column: 0,1,2,3…; `weight` = bin support).
- In .call.cns, `log2` ±0.3 is the usual “gain/loss” noise floor for WES; arm-level events in HGSOC (chr8q gain, chr1q gain, chr16/17 loss, PTEN/RB1 loss) are the biologically expected pattern — and the corrected full-round run indeed found 9q single-copy loss, chr8 gain (cn=3, spanning MYC), chr20/21 gains and chrY loss at high weight (§ 10.2).
- `--diagram` produces the genome diagram PDF; `--scatter` the per-chromosome log2 scatter. CNVkit’s `metrics` and `segmetrics` give per-segment robust stats.
- Sex chromosomes: male/female mismatch between tumor and normal inflates chrX/Y — CNVkit normally handles it if sex is declared (`--sex`); check the log.

10.5 Tuning & beyond 调优与扩展

- Purity/ploidy: `cnvkit.py call --purity 0.5` integrates with the TMB/AF interpretation.
- `cnvkit.py export seg/vcf/theta` bridges to DNAseq tools, IGV, ABSOLUTE/THetA2.
- GISTIC on a cohort of .cns/seg files → recurrent amplifications/deletions; that is the “cohort version” of this step.
- For immune-oncology course tie-ins: CNV of PD-L1 (CD274), copy-loss of HLA locus, and CCNE1 amplification are the classic HGSOC/immuno-evasion stories to grep for in the output.

11. Interpretation checklist · 结果解读检查清单

The original roadmap closes with an 11-point checklist. Here it is again — now with **actual answers from both runs**, and the one missing metric (on-target depth) added.

#	Question	Demo (0.5×) answer	Full (~55×/91× usable on-target) answer
1	Do FASTQ reports pass Q30 thresholds?	Yes: 97.0% / 97.8% (post-trim)	Yes: 97.0% / 97.8% (post-trim; raw 95.5%/96.9%)
2	Is mapping rate >95%?	100%	100% — but note: this **dataset ships pre-filtered** (only 311/79.6M unmapped in the full round); fresh lab data is 96–99%
3	Properly paired >90%?	OC 83.3%, PBMC 99.5%	OC **83.2%** (real tumor discordance — 15.8% mates on other chr), PBMC 99.4%
4	Duplication <20%?	~0.03–0.06% (subsample artifact)	OC **7.08%**, PBMC **10.77%** (teacher’s: 7.39/11.01)
5	**On-target depth sufficient?** (the metric the roadmap forgot)	**0.3×/0.5× usable, median 0 — NO**	OC **54.6×** (median 57; 87.4% ⩾30×), PBMC **91.3×** (median 96; 89.6% ⩾30×) — borderline-adequate tumor, good normal
6	Insert size within capture design range?	peak **150 bp** (short!)	150 bp — 25% of usable bases lost to mate overlap (HsMetrics `PCT_EXC_OVERLAP`)

#	Question	Demo (0.5×) answer	Full (~55×/91× usable on-target) answer
7	Are known sites recovered (dbsnp overlap)?	7,388 truth sites	**28,324** truth sites (teacher's genome-wide: 85,159)
8	Ti/Tv of PASS variants?	joint 2.59; PASS 2.60 — novel Ti/Tv **hugs** known (2.33 vs 2.65): no discrimination	PASS **2.48** (known 2.49) — healthy range
9	Het/Hom ratio ~1.5 for exome?	**0.14** — genotypes collapse at 0.5× (single-read "homs")	**1.45** — healthy
10	Somatic count plausible for tumor type?	0 (no power)	**108 PASS / TMB 1.78 mut/Mb** — squarely in HGSOC range
11	Do CNV calls make karyotypic sense?	No — pure noise (§ 10.2)	**Yes** (corrected hybrid mode): 9q cn=1 loss, chr8 gain cn=3, chrY loss — coherent HGSOC karyotype

The missing metric — add this to every WES run (issue #9):

```
# CollectHsMetrics wants a Picard interval_list with a SAM header — convert the BED first
gatk --java-options "-Xmx4g" BedToIntervalList \
  -I "$TARGETS" -O "$OUT/qc/targets.interval_list" -SD "$REF/hg38.dict"

gatk --java-options "-Xmx8g" CollectHsMetrics \
  -I "$OUT/1_alignment/OC/OC_sorted.markdup.BQSR.bam" \
  -O "$OUT/1_alignment/OC/OC.hs_metrics.txt" \
  --TARGET_INTERVALS "$OUT/qc/targets.interval_list" \
  --BAIT_INTERVALS "$OUT/qc/targets.interval_list" \
  --REFERENCE_SEQUENCE "$REFERENCE"   # no separate bait list on the cluster; targets≈baits for
teaching
```

(Two practical notes learned the hard way in this very round: plain BEDs are rejected — *`Interval list file must contain header`* — hence the BedToIntervalList step; and the sequence dictionary lives at *`hg38.dict`*, not *`hg38.fa.dict`*.)

The rows that matter: `MEAN_TARGET_COVERAGE`, `PCT_TARGET_BASES_30X` (or 50X/100X columns), `PCT_SELECTED_BASES` (on-target + near-bait = capture specificity), `FOLD_ENRICHMENT`, and — surprisingly informative — `PCT_EXC_OVERLAP` / `PCT_EXC_DUPE` / `PCT_EXC_OFF_TARGET`: the three places your paid-for depth disappears.

Measured, both rounds (all four files under `wes_run_full_20260920/qc/`):

Sample	MEAN_TARGET_COV	⩾30X	⩾100X	PCT_SELECTED	ZERO_CVG	EXC_DUPE	EXC_OVERLAP
demo OC	0.3×	0%	0%	79.9%	59.4%	0.03%	26.8%
demo PBMC	0.5×	0%	0%	90.8%	46.6%	0.06%	24.9%
full OC	**54.6×**	87.4%	74.2%	79.9%	8.4%	7.3%	25.6%
full PBMC	**91.3×**	89.6%	87.0%	90.7%	8.1%	11.0%	23.0%

中文批注：检查清单的 11 个问题里，最容易被"绕过去"的是 #5（靶区深度够不够）——因为整条路线图里没有任何命令真正测量过它。上表就是实测结果：全量轮肿瘤 54.6×、正常 91.3×，80–100× 的临线

刚好够用；demo 轮中位深度是 **0**。另外注意两个实测细节：① 这套文库插入片段峰值只有 150bp，双端重叠吃掉约 25% 的可用碱基——这是”付了 100× 的钱、拿到 55× 的深度”的主要原因；② 零覆盖靶区 8.4%——AllTracks 版 BED 含大量难捕获区域，这属于捕获设计的固有属性，不是实验失败。

12. File summary · 产物清单

What each run directory contains when everything is done (paths as produced by the verbatim pipeline; sizes from the full round, demo sizes in parentheses where notable):

Artifact (full round path under `wes_run_full_20260920/`)	Size	Notes
`0_fastq/OC/OC.fastp.html/.json` (+ clean FASTQs)	~0.5 G clean FASTQ pairs	Q30 97.0% post-trim
`1_alignment/OC/OC_sorted.bam`	4.0 GB	piped `bwa|sort` — no 33-GB SAM ever existed (issue #7)
`1_alignment/OC/OC_sorted.markdup.bam`	4.7 GB	1. `.bai` via `--CREATE_INDEX`
`1_alignment/OC/OC_sorted.markdup.BQSR.bam`	8.8 GB	~1.9× the markdup BAM — recompression quirk; budget for it
`1_alignment/OC/OC.g.vcf`	209 MB	GVCF per sample (PBMC: 59 MB)
`2_variant_call/combined.g.vcf`	264 MB	2-sample combined GVCF
`2_variant_call/OC_blood_variants.vcf`	14 MB	**55,123 records** — the raw joint callset
`2_variant_call/OC_blood.snps.recal` + `.tranches` (+indel)	4.3 MB + 0.6 MB	VQSR models
`2_variant_call/OC_blood.all.VQSR.vcf`	30 MB	**110,246 records — the §7.3 duplication, as-written**
`2_variant_call/OC_blood.all.sequentialVQSR.vcf`	16 MB	**55,123 records — the corrected output**
`3_annotation/OC_blood_anno.hg38_multianno.txt`	10 MB	basic protocol, on the duplicated VCF (110,284 rows)
`3_annotation/OC_blood_anno_rich_installed.hg38_multianno.txt`	27 MB	**corrected rich protocol: 55,165 rows × 146 cols**
`5_somatic_tmb/somatic.raw.vcf.gz` → `somatic.filtered.vcf.gz` → `somatic.pass.vcf.gz`	0.3 → 0.35 → 0.02 GB	2,949 raw → 108 PASS
`4_cnv/cnvkit_out/` (amplicon, verbatim)	~47 MB	empty antitarget files — the tell (§10.2)
`4_cnv/cnvkit_out_hybrid/` (corrected)	~55 MB	real antitarget bins; `--scatter`/`--diagram` PDFs in both
`qc/*.hs_metrics.txt`	4 files	the §11 coverage numbers, both rounds

(Demo round: same layout, everything ~50–100× smaller — e.g. joint VCF 2.8 MB, BQSR BAM 54 MB.)

13. Summary · 总结与两轮对比

What the two rounds taught, in one table:

Aspect	Demo round (0.5×)	Full round (54.6×/91.3× usable on-target)
Purpose	Pipeline mechanics	Real biology
Alignment	100% mapped (dataset pre-filtered), OC properly-paired 83.3%	identical profile: OC 83.2% (real tumor discordance), PBMC 99.4%
Germline joint callset	14,782 records	**55,123 records** (2-sample, on-target; teacher's genome-wide 1-sample: 204,178)
VQSR	ran; truth sites 7,388; minVQSLod@100% = −5889; novel Ti/Tv hugs known (2.33 vs 2.65)	truth sites 28,324; tranches behave like a real (if thin) model; sequential correction removes 2.1%
§ 7.3 duplication bug	29,564 = 2 × 14,782	**110,246 = 2 × 55,123** (corrected: 55,123, zero unfiltered twins)
ANNOVAR	basic OK, rich protocol fails (missing DBs)	verbatim rich still fails; **corrected names: 55,165 rows × 146 cols** incl. REVEL/CADD/AlphaMissense
Somatic / TMB	0 — no statistical power	**108 PASS / TMB 1.78 mut/Mb** (HGSOC-plausible)
CNVkit	cn=6345 absurdities	verbatim amplicon: empty antitarget, muted events; **corrected hybrid: 9q loss, chr8 gain — coherent HGSOC karyotype**
Wall time	~25 min (1 job)	3 h 42 m longest chain (8 jobs, dependency-linked; prep 2:20+3:42 in parallel)

The four lessons to carry away:

1. **"It ran" ≠ "it's right."** The VQSR duplication bug (§ 7.2) and the CNVkit mode error (§ 10.3) both run to completion. The single most important skill this course builds is counting your own outputs (variants, rows, sites) and knowing what numbers are plausible.
2. **Depth gates everything.** 0.5× is enough to exercise the machinery, and it is honest about its limits (somatic=0, CNV=noise). Interpretation belongs to the full-depth round.
3. **Tools encode assumptions** (capture mode, sample naming, database names, "VQSR needs 30 samples"). Reading the manual after the first error is normal; reading it before the first run is professional.
4. **Corrected, the pipeline is a strong teaching skeleton**: the roadmap's step order, RG discipline, GVCF workflow, and germline-vs-somatic separation are all best-practice — the fixes are surgical, not structural.

中文批注：两轮对比最大的价值是把"流程问题"和"数据问题"彻底分开：demo 轮暴露的 0、噪声、荒谬 CN 值是**数据深度问题**（换全量数据自然消失）；VQSR 重复、ANNOVAR 库名、CNVkit 模式是**流程本身的问题**（换什么数据都在，连老师的参考结果里都有）。把这两类问题分清楚，你就掌握了"排错"的第一层功力。

Reproduction · 复现说明

Both rounds were run as SLURM jobs (partition `cn-long`, QoS `bjx131cnl`, account `bjx131_g1`, `--mem=0`, 8–16 cores per job):

Round	Directory	Scripts
demo	`/lustre1/user/bjx131_pkuhpc/wes_run_cluster_20260914`	`scripts/wes_demo_steps3to10.sbatch` (fastp was already done in a prior session)
full (verbatim)	`/lustre1/user/bjx131_pkuhpc/wes_run_full_20260920`	`scripts/full_prep_OC.sbatch`, `scripts/full_prep_PBMC.sbatch`, `scripts/full_joint.sbatch`, `scripts/full_somatic.sbatch`, `scripts/full_cnv.sbatch` (joint/somatic/cnv submitted with `--dependency=afterok:<prep jobs>`)
full (corrected extras)	same	`scripts/full_hsmetrics.sbatch` (§11), `scripts/full_cnv_hybrid.sbatch` (§10.3), `scripts/full_fix_vqsr.sbatch` (sequential ApplyVQSR + rich ANNOVAR with installed DB names)

Each run directory contains `log/deviations.log` (every expected-failure and deviation with timestamps) and `log/summary_*.log` (the numbers quoted throughout this edition). The full round keeps the roadmap's literal SAM→BAM→sort chain only in the demo round; the full round used the piped variant (issue #7, logged).

Roadmap fidelity statement: every command in steps 3–10 was executed exactly as printed in the roadmap (same flags, same file names, same order). The only wrapper additions are: SLURM headers, environment activation, preflight existence checks, expected-failure `if/else` guards (which log-and-continue instead of dying), and the summary blocks. The single algorithmic deviation is the `bwa | sort` pipe in the full round, which changes no output content.

Appendix A. Issue registry · 问题清单（完整版）

Each issue: **where** (roadmap section), **what** (symptom), **evidence** (measured), **fix** (what this edition recommends), **severity**.

A1. VQSR MergeVcfs duplication — Major

- Where: §7.3. What: `OC_blood.all.VQSR.vcf` contains every variant twice.
- Evidence: demo 29,564 records / 14,782 unique sites; our full round 110,246 / 55,123; teacher 408,356 / 204,178. Mechanism: ApplyVQSR (either mode) emits the whole callset (the other variant type passes through with FILTER `.`); MergeVcfs concatenates two whole callsets. Each site ends up as one SNP-verdict record + one `.` record — the PASS count happens to stay right, which masks the bug during casual QC.
- Fix: sequential ApplyVQSR (SNP → INDEL), or SelectVariants split → filter → merge. Measured result of the fix (job 3056): 55,123 records / 55,123 unique sites / 0 unfiltered twins; filter rate 2.1%.
- Severity: Major — silently corrupts every downstream count and annotation.

A2. ANNOVAR rich protocol names unavailable — Major

- Where: §8.3. What: `dbnsfp30a`, `exac03`, `gnomad_genome`, `avsnp147` absent from `/lustre1/share/annovar_new/humandb` (which has `dbnsfp54a`, `gnomad211_exome`, `gnomad41_exome`, `avsnp151`).
- Evidence: table_annovar fails in both runs; `ls humandb` listing.
- Fix: use the installed names (§8.3). Severity: Major (hard failure).

A3. VQSR on an undersized callset — Major (statistical)

- Where: § 7. What: with 2 exomes / ~15k variants (demo) the Gaussian mixture is unidentifiable; tranches become meaningless.
- Evidence (all three callsets measured, § 7.2): truth sites 7,388 (demo) vs 28,324 (our full) vs 85,159 (teacher); minVQSLod at the 100% tranche −5889.9 (demo) vs −1111 (full) vs −25.5 (teacher); and the decisive pattern: healthy callsets show novel Ti/Tv well below known (teacher: 1.69 vs 2.34) while the demo's novel Ti/Tv **hugs** known (2.46 vs 2.71) — a model that cannot separate its worst from its best variants.
- Fix: for ⩽10 samples use VariantFiltration hard filters (§ 7.3); VQSR is a cohort tool (≈30+ samples). Severity: Major — silently wrong filtering.

A4. CNVkit `-m amplicon` on a hybrid-capture panel — Major

- Where: § 10.2. What: S07604514 is a SureSelect V6 hybrid capture; amplicon mode discards all off-target reads and builds no antitarget baseline.
- Evidence (both rounds measured): demo produced cn up to 6345 and 169-Mb "losses" at 0.5×; the full-round verbatim run produced **empty antitarget coverage files (31 bytes)** and muted arm-level events, while the corrected hybrid run on the same BAMs found 9q single-copy loss, chr8 gain (cn=3), chrY loss at high weight (§ 10.2). The pipeline completes either way.
- Fix: `-m hybrid` with `-g access-5kb.hg38.bed` (both on the cluster). Severity: Major — results unusable as-is.

A5. Mutect2 without a Panel of Normals — Medium

- Where: § 9.1. What: PoN filters recurring capture/alignment artifacts; the cluster ships `1000g_pon.hg38.vcf.gz` but the roadmap doesn't use it.
- Fix: `--pon $REF/1000g_pon.hg38.vcf.gz`. (Also: gnomAD AF-only is the recommended germline resource; 1000G high-conf works but is dated.)

A6. Dataset path inconsistency — Medium

- Where: § 0.2, README, old vs new roadmap. What: three different `$DATA` targets (`OC_WES` deprecated downsample / `data_new` new downsample / an **empty** `analysis/0_preprocess` in the old HTML).
- Fix: this edition pins every dataset by absolute path with role labels (§ 0.2).

A7. 33 GB intermediate SAM per sample — Medium (efficiency)

- Where: § 3.1–3.3. Fix: `bwa | sort` pipe (full round). Output identical; disk and ~30–40% wall time saved. (Demo round kept the literal chain for fidelity.)

A8. TMB counts source inconsistency — Minor

- Where: § 9.2/9.3: `somatic.pass.vcf.gz` written, but TMB counted from `somatic.filtered.vcf.gz` PASS subset. Same number; keep one convention.

A9. No on-target depth metric — Minor

- Where: § 11 checklist asks, no command measures. Fix: CollectHsMetrics (§ 11).

A10. Naming inconsistencies — Minor

- `SM:blood` vs "PBMC" directories (functional but confusing); teacher's `2_vairant_call` directory typo (reference side); `vcf4old` format flag in ANNOVAR (works; `vcf4` is the modern name).

A11. Runtime environment quirks (not roadmap bugs, but you will hit them) — Info

- SLURM on this cluster: `--account=bjx131_g1 --qos=bjx131cnl` mandatory; `--mem=0` required (RealMemory=1 misconfiguration); conda activate before `set -u`.
- GATK bundle `.idx` files all present — but check on any new cluster.

- CollectHsMetrics wants a Picard interval_list (run BedToIntervalList on the BED first) and the dictionary is `hg38.dict`, not `hg38.fa.dict`.

A12. The teacher's reference result deviates from the printed roadmap — Major (comparability)

- Where: `analysis_full/2_vairant_call` (the reference everyone diffs against).
- Evidence (read from the VCF headers and the files themselves):
  1. **Single-sample "joint" call:** `combined.g.vcf` has only sample `OC` --- the blood gVCF's RG `SM` tag was set to `OC` (sample-name collision), so CombineGVCFs collapsed both samples into one. The printed roadmap clearly intends a 2-sample joint call (OC + blood).
  2. **No `-L` anywhere:** HaplotypeCaller/GenotypeGVCFs ran genome-wide — 72% of the 204,178 sites lie outside the capture targets, many with shallow, low-quality coverage.
  3. §7.3′s MergeVcfs duplication is present in the reference output as well (408,356 = 2 × 204,178 — issue A1).
- Consequence: "204,178 vs my 55,123" is **not** a depth comparison — it's (genome-wide × 1 sample × doubled by the merge bug) vs (on-target × 2 samples). Diff callsets only after reading their `##GATKCommandLine` headers.
- Lesson for the course: the RG `SM` tag discipline the roadmap itself preaches (§2) is exactly what silently bit its own reference run. This is the best argument for step 0′s "check your BAM header" habit.
- Fix: nothing to fix in the roadmap text (it prescribes `-L` and correct SM tags); this entry exists so students comparing against `analysis_full` know what the differences are and why.

Appendix B. One-page corrected pipeline · 一页矫正版命令速查

```
# 0) env
source /lustre1/share/miniconda3/etc/profile.d/conda.sh && conda activate wes
export SHARE=/lustre1/share REF=$SHARE/references
export TARGETS=$REF/S07604514_AllTracks_V6_60_hg38.bed REFERENCE=$REF/hg38.fa
export DBSNP=$REF/dbsnp_138.hg38.vcf INDELS=$REF/Mills_and_1000G_gold_standard.indels.hg38.vcf
export HIGHCONF=$REF/1000G_phase1.snps.high_confidence.hg38.vcf
export PON=$REF/1000g_pon.hg38.vcf.gz ACCESS=$REF/access-5kb.hg38.bed
export ANNOVAR=$SHARE/annovar_new HUMANDB=$ANNOVAR/humandb

# 2) QC
fastp -i R1.fq.gz -I R2.fq.gz -o R1.clean.fq.gz -O R2.clean.fq.gz \
  --detect_adapter_for_pe --cut_right --cut_right_window_size 4 --cut_right_mean_quality 20 \
  --qualified_quality_phred 20 --unqualified_percent_limit 40 --n_base_limit 5 \
  --length_required 50 --thread 8 --html S.html --json S.json

# 3) align (piped)
bwa mem -t 8 -M -R "@RG\tID:S\tPL:illumina\tSM:S\tLB:libS" $REFERENCE R1.clean.fq.gz R2.clean.fq.gz
\
  | samtools sort -@ 8 -m 4G -o S_sorted.bam -
samtools index -@ 8 S_sorted.bam
samtools flagstat -@ 8 S_sorted.bam > S.flagstat.txt
gatk BedToIntervalList -I $TARGETS -O targets.interval_list -SD $REF/hg38.dict
gatk CollectHsMetrics -I S_sorted.bam -O S.hs_metrics.txt \
  --TARGET_INTERVALS targets.interval_list --BAIT_INTERVALS targets.interval_list \
  --REFERENCE_SEQUENCE $REFERENCE   # NEW: on-target depth (§11)

# 4) markdup
gatk MarkDuplicates -I S_sorted.bam -O S_sorted.markdup.bam -M S_markdup_metrics.txt --CREATE_INDEX
true

# 5) BQSR
gatk BaseRecalibrator -R $REFERENCE -I S_sorted.markdup.bam -L $TARGETS \
  --known-sites $DBSNP --known-sites $INDELS --known-sites $HIGHCONF -O S.recal.table
gatk ApplyBQSR -R $REFERENCE -I S_sorted.markdup.bam --bqsr-recal-file S.recal.table -O S.BQSR.bam
```

```
# 6) germline GVCF workflow
gatk HaplotypeCaller -ERC GVCF -R $REFERENCE -I S.BQSR.bam -D $DBSNP -L $TARGETS \
  --native-pair-hmm-threads 8 -O S.g.vcf
gatk CombineGVCFs -R $REFERENCE -V OC.g.vcf -V blood.g.vcf -O combined.g.vcf
gatk GenotypeGVCFs -R $REFERENCE -V combined.g.vcf -D $DBSNP -O joint.vcf

# 7) filtering — small callset: hard filters (VQSR needs ~30+ samples)
gatk VariantFiltration -V joint.vcf -O joint.filtered.vcf \
  --filter-name SNP_QD2 --filter-expression "QD < 2.0 && SNP" \
  --filter-name SNP_MQ40 --filter-expression "MQ < 40.0 && SNP" \
  --filter-name SNP_FS60 --filter-expression "FS > 60.0 && SNP" \
  --filter-name SNP_SOR3 --filter-expression "SOR > 3.0 && SNP" \
  --filter-name INDEL_QD2 --filter-expression "QD < 2.0 && INDEL" \
  --filter-name INDEL_FS200 --filter-expression "FS > 200.0 && INDEL" \
  --filter-name INDEL_SOR10 --filter-expression "SOR > 10.0 && INDEL"
# (cohort scale instead: VariantRecalibrator SNP → ApplyVQSR → VariantRecalibrator INDEL
#  → ApplyVQSR on the SNP output — sequential, never MergeVcfs of two whole callsets)

# 8) ANNOVAR (installed databases)
perl $ANNOVAR/convert2annovar.pl -format vcf4 joint.filtered.vcf -includeinfo -comment \
  -out ann.avinput
perl $ANNOVAR/table_annovar.pl ann.avinput $HUMANDB -buildver hg38 -out anno \
  -remove -protocol refGeneWithVer,cytoBand,gnomad41_exome,dbnsfp54a,avsnp151 \
  -operation g,r,f,f,f

# 9) somatic TMB (PoN added)
gatk Mutect2 -R $REFERENCE -I OC.BQSR.bam -I blood.BQSR.bam --normal-sample blood \
  -L $TARGETS --germline-resource $HIGHCONF --pon $PON \
  -O somatic.raw.vcf.gz --native-pair-hmm-threads 8 --f1r2-tar-gz f1r2.tar.gz
gatk LearnReadOrientationModel -I f1r2.tar.gz -O ob.tar.gz
gatk FilterMutectCalls -R $REFERENCE -V somatic.raw.vcf.gz --stats somatic.raw.vcf.gz.stats \
  --ob-priors ob.tar.gz -O somatic.filtered.vcf.gz
TMB=$(bcftools view -H -f PASS somatic.filtered.vcf.gz | wc -l)
echo "TMB = $(awk -v n=$TMB -v mb=60.5079 'BEGIN{printf "%.2f", n/mb}') mut/Mb"

# 10) CNVkit (hybrid mode)
cnvkit.py batch OC.BQSR.bam -n blood.BQSR.bam -f $REFERENCE -t $TARGETS \
  -g $ACCESS -m hybrid --output-reference reference.cnn -d cnvkit_out \
  --scatter --diagram -p 8
```

Generated from the two-run reproduction described in § 13; all quoted numbers are measured in those runs. Chinese annotations target the course's bilingual audience.